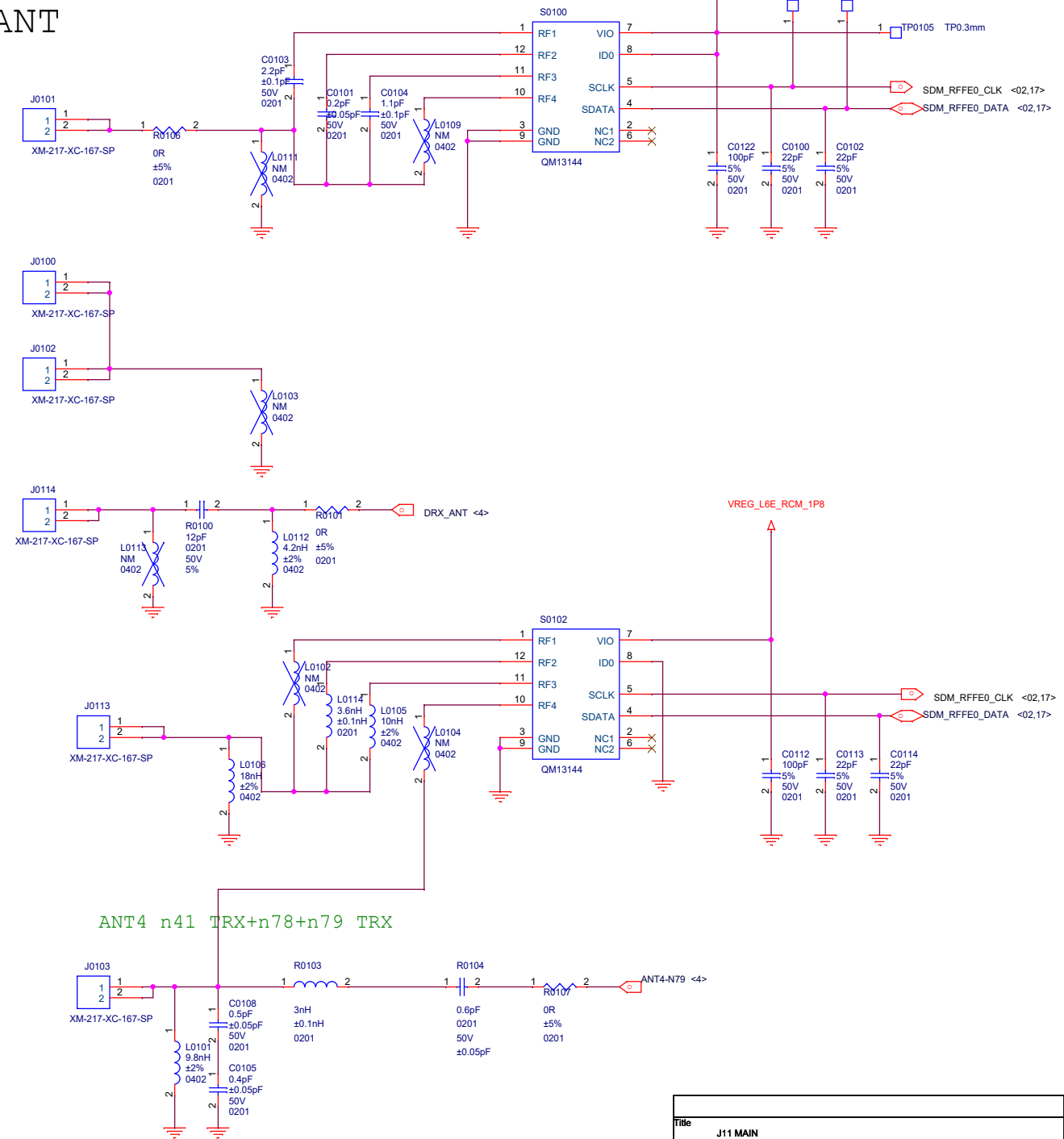
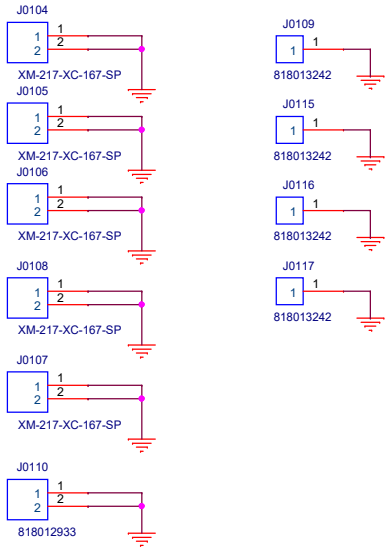


DRX_ANT

接地弹片 接地垫片 夹子

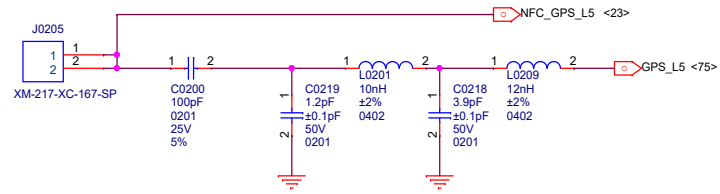


ANT4 n41 TRX+n78+n79 TRX

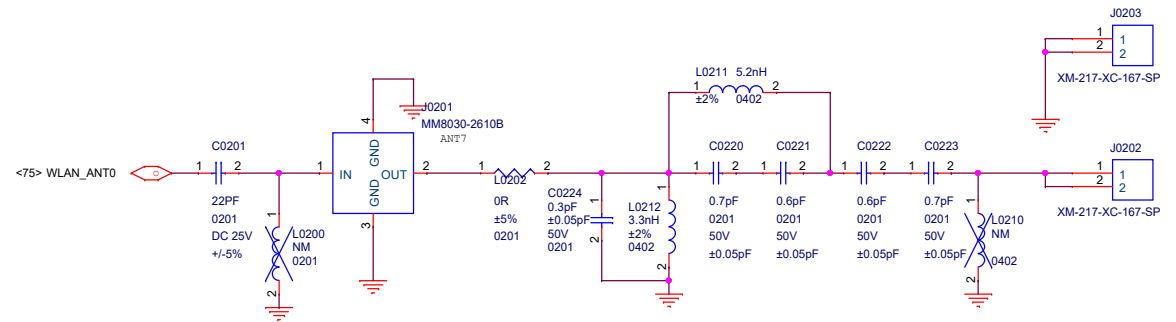
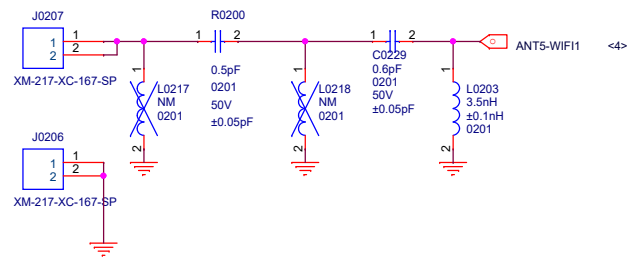
Title		J11 MAIN
Size	A3	Document Number
Date:		Monday, December 23, 2019
Sheet		02 of 74

GPS L1+ WIFI2.4G CH0+WIFI 5G ANT2

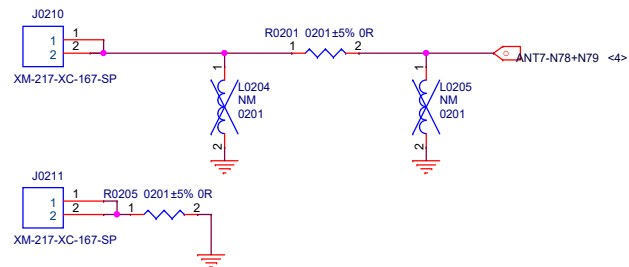
GPS L5



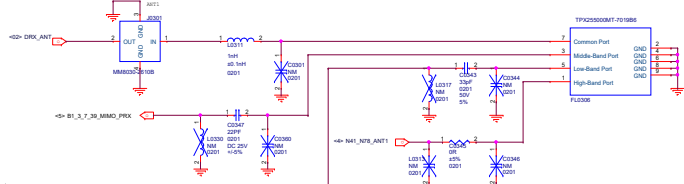
ANT5 WIFI2.4G+5G



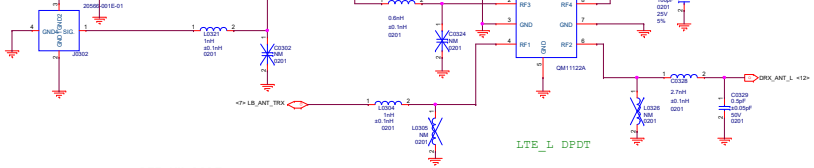
ANT7 N78+N79



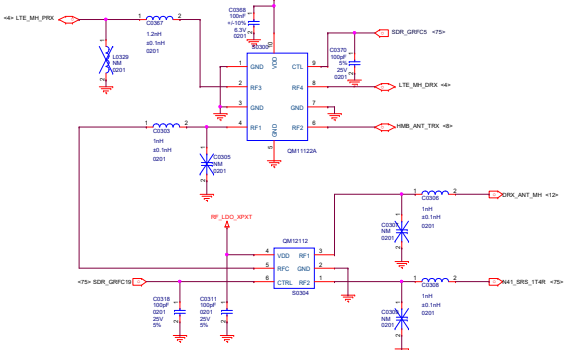
2/3/4G+n41+n79 ANT1



2/3/4G+n41+n79 ANT0

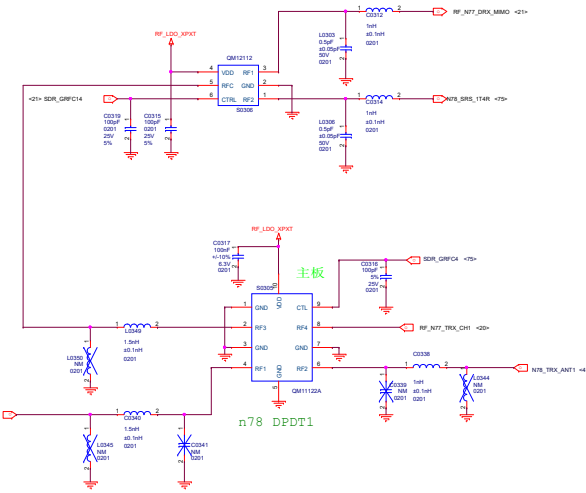
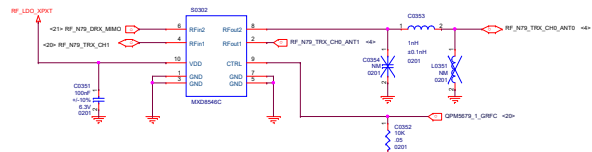


LTE_MH DPDT 主板



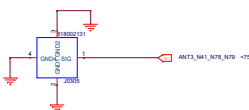
LTE_L DPDT

N79 DPDT2 主板

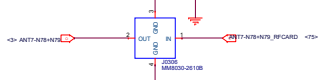


N78 ANT8

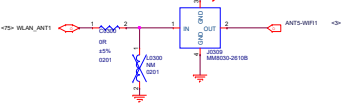
n41 TRX+n78 TRX+n79 TRX ANT3



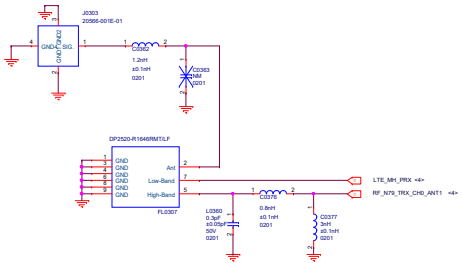
N78 N79 ANT7



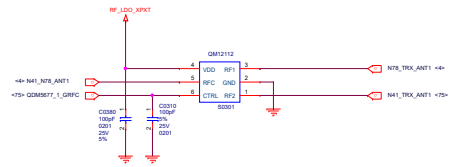
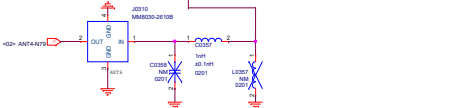
WIFI2.4G+WIFI 5G CH1 ANT5



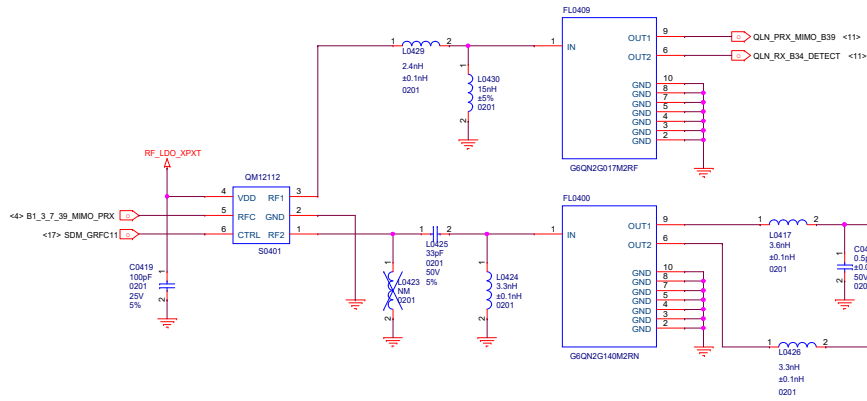
N41 TRX+n78 ANT9



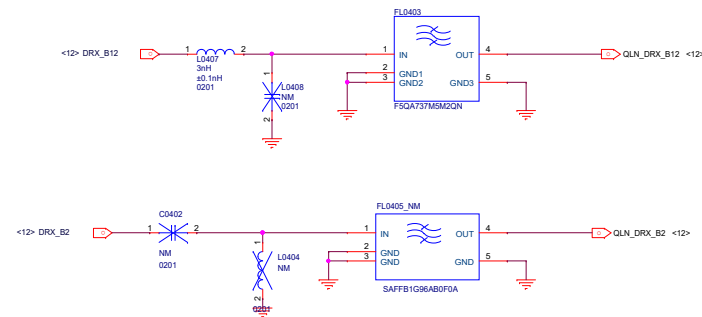
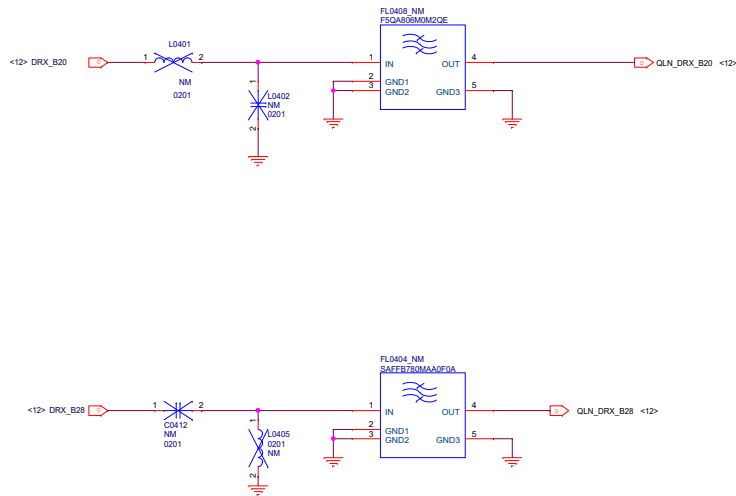
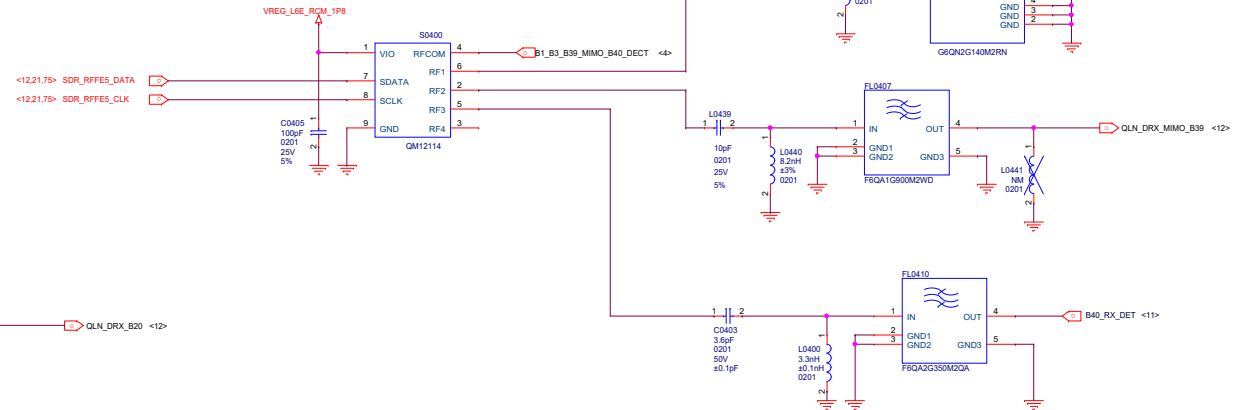
n41 TRX+n78+n79 TRX ANT4



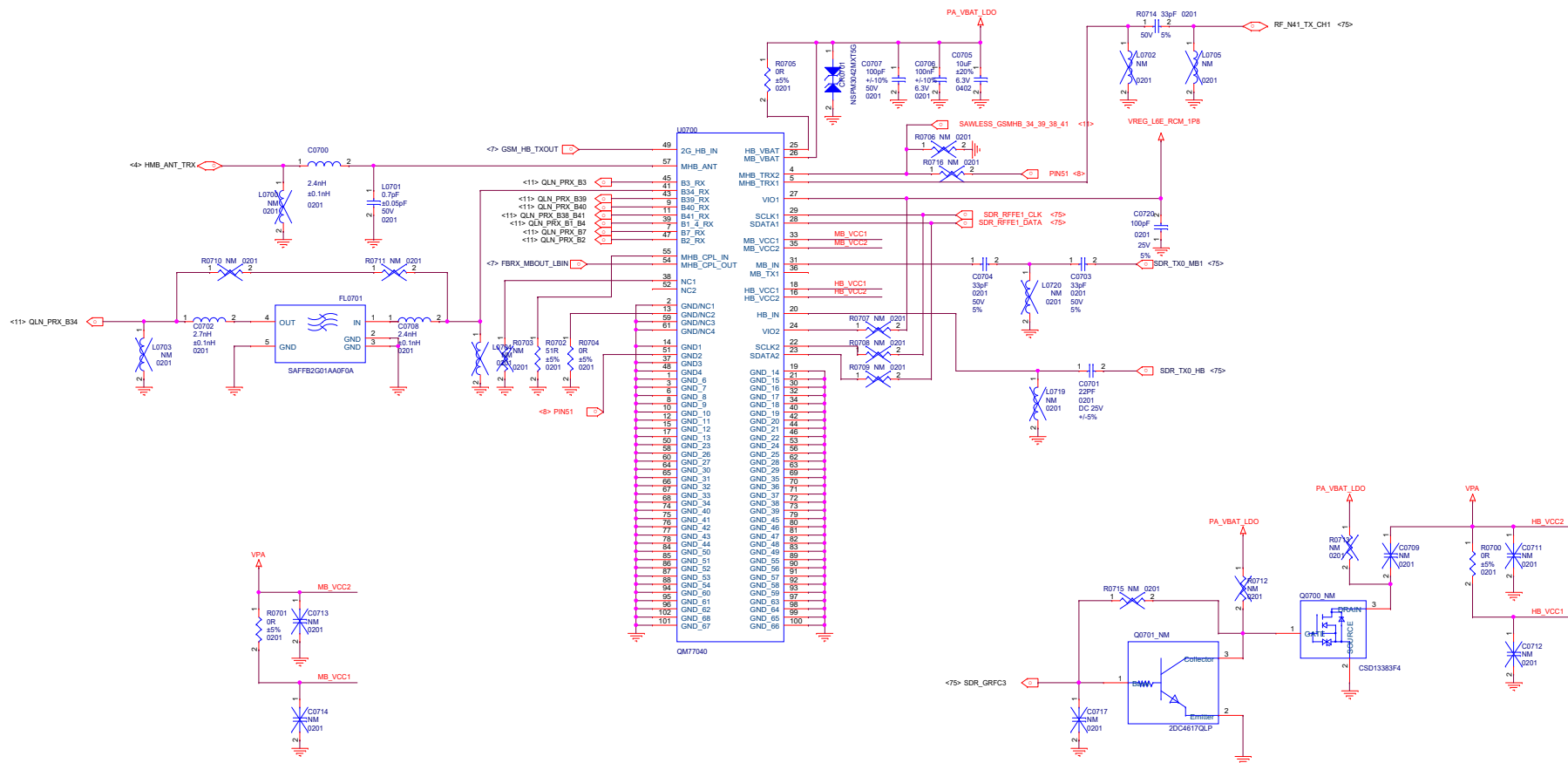
B1+3+7+39 MIMO PRX



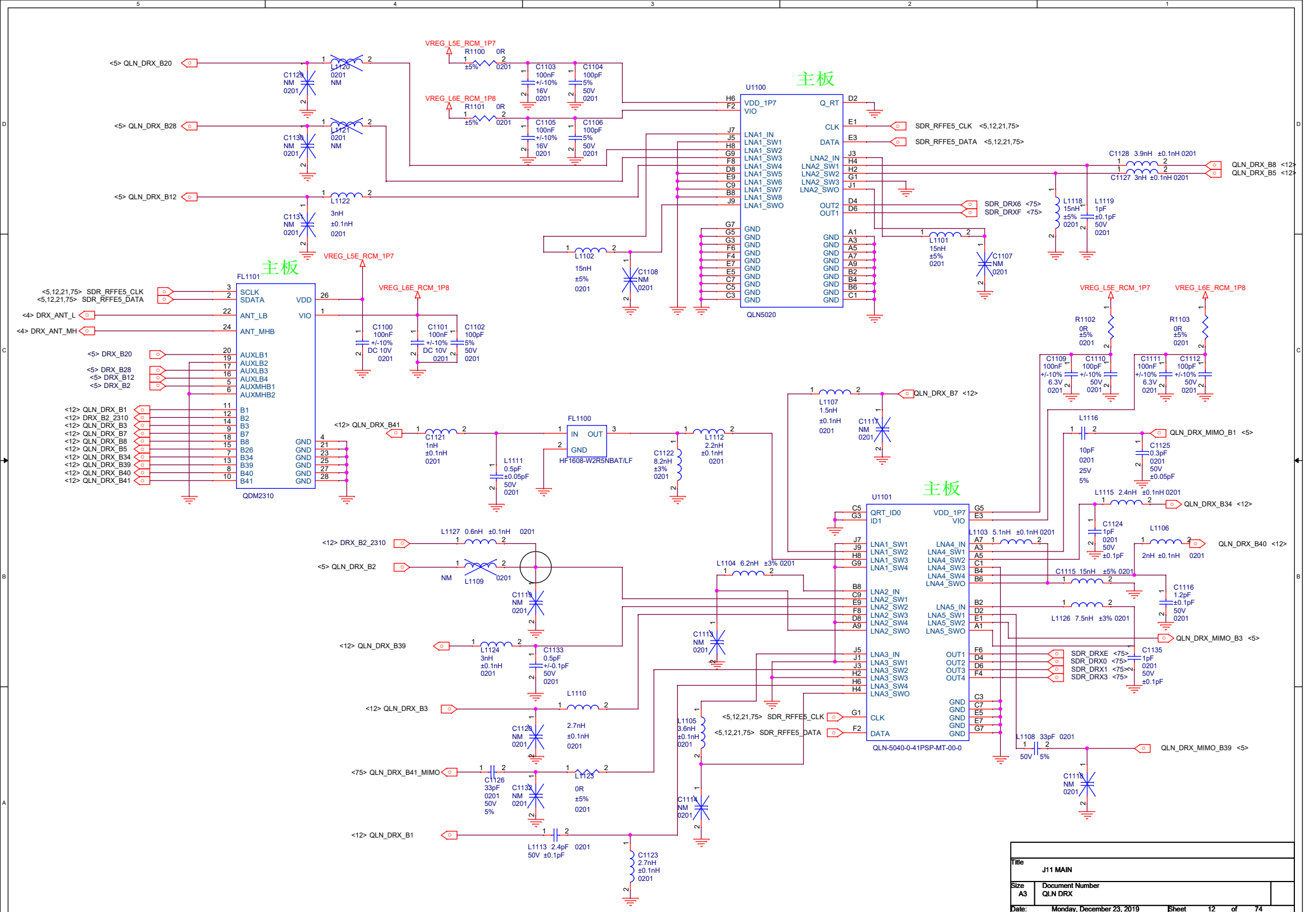
B1+3+7+39 MIMO DRX



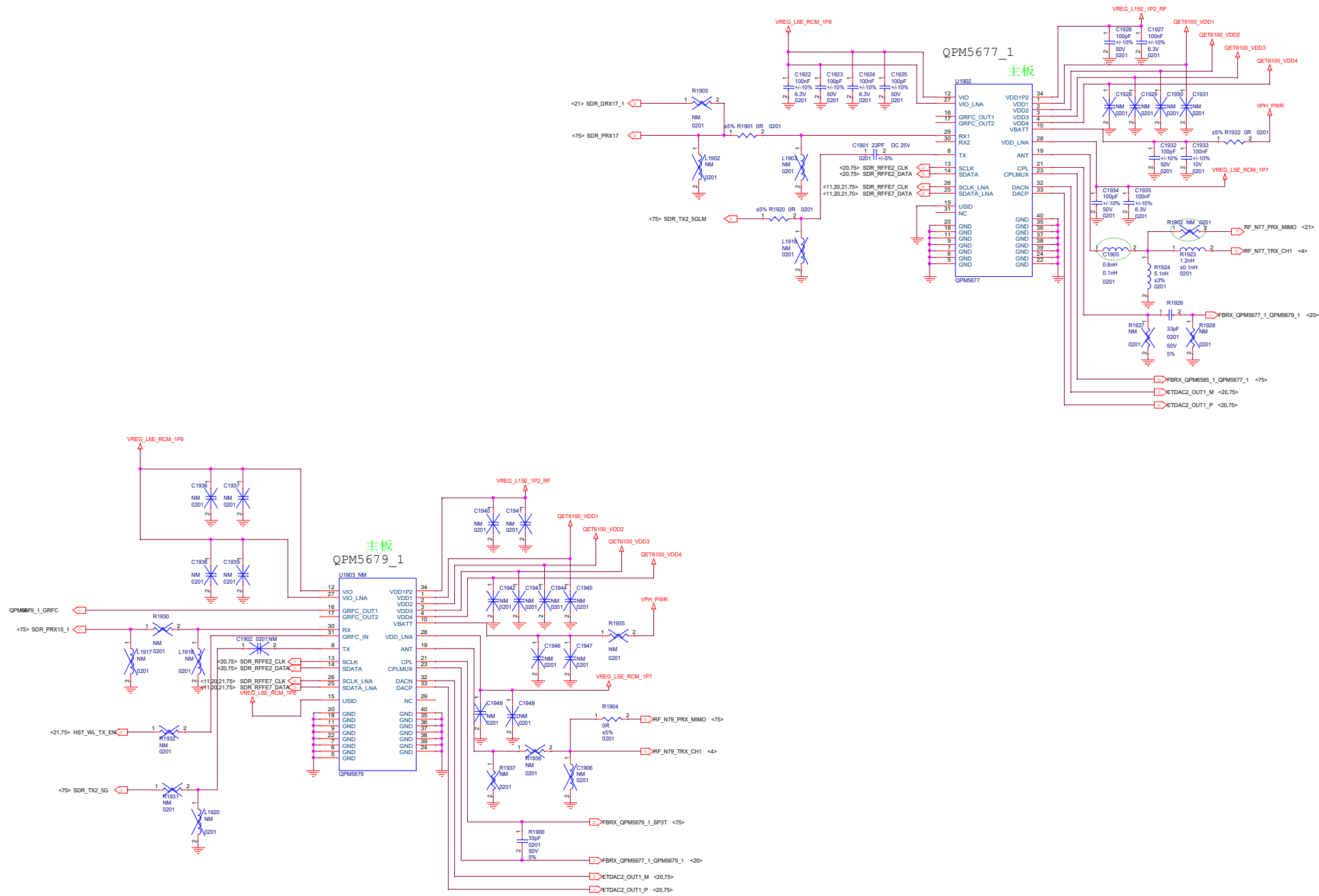
Title	J11 MAIN
Size	Document Number
A2	<Doc>
Date	Monday, December 23, 2019



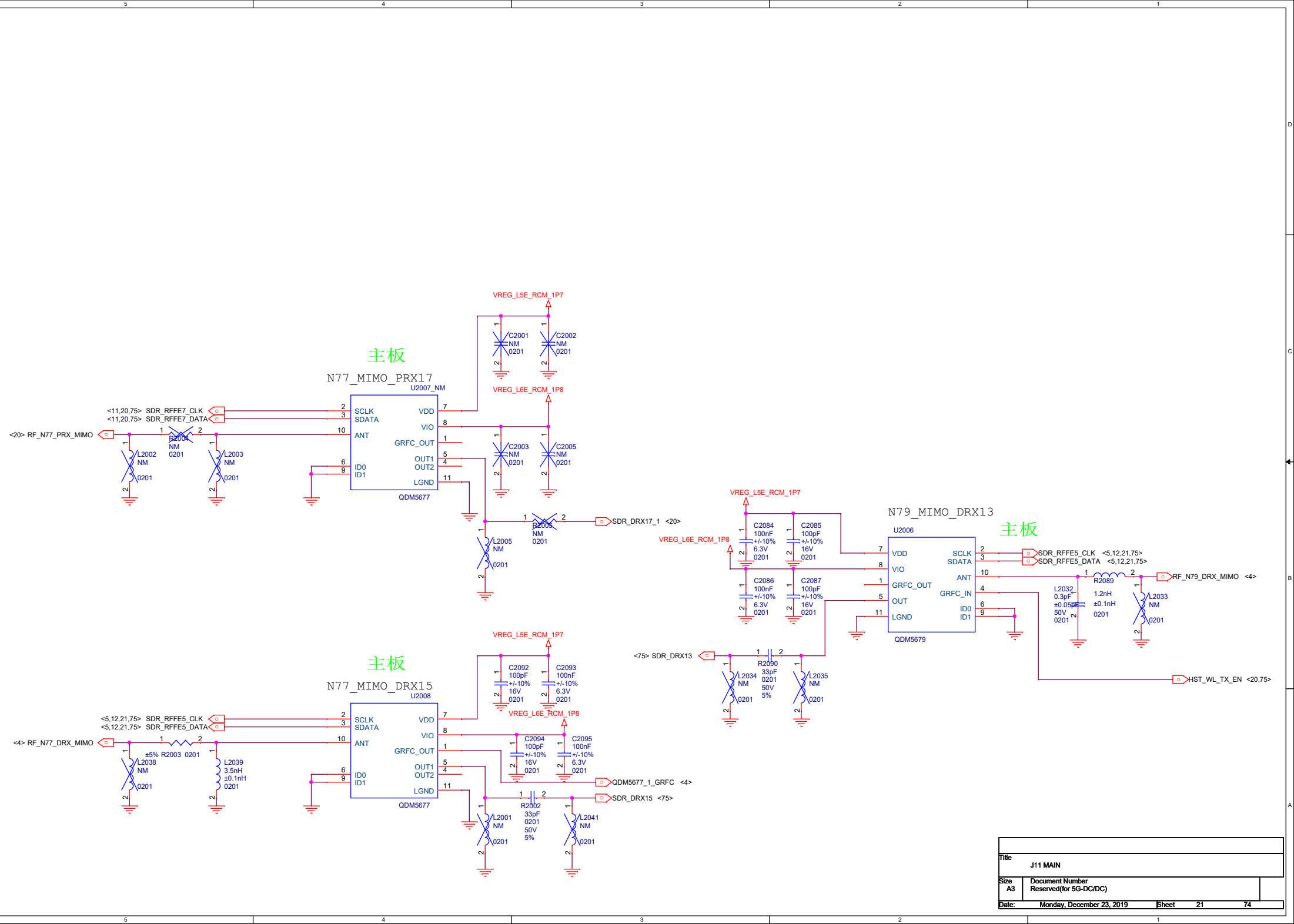
Title J11 MAIN		
Size A2	Document Number PA2	
Date:	Monday, December 23, 2019	Sheet

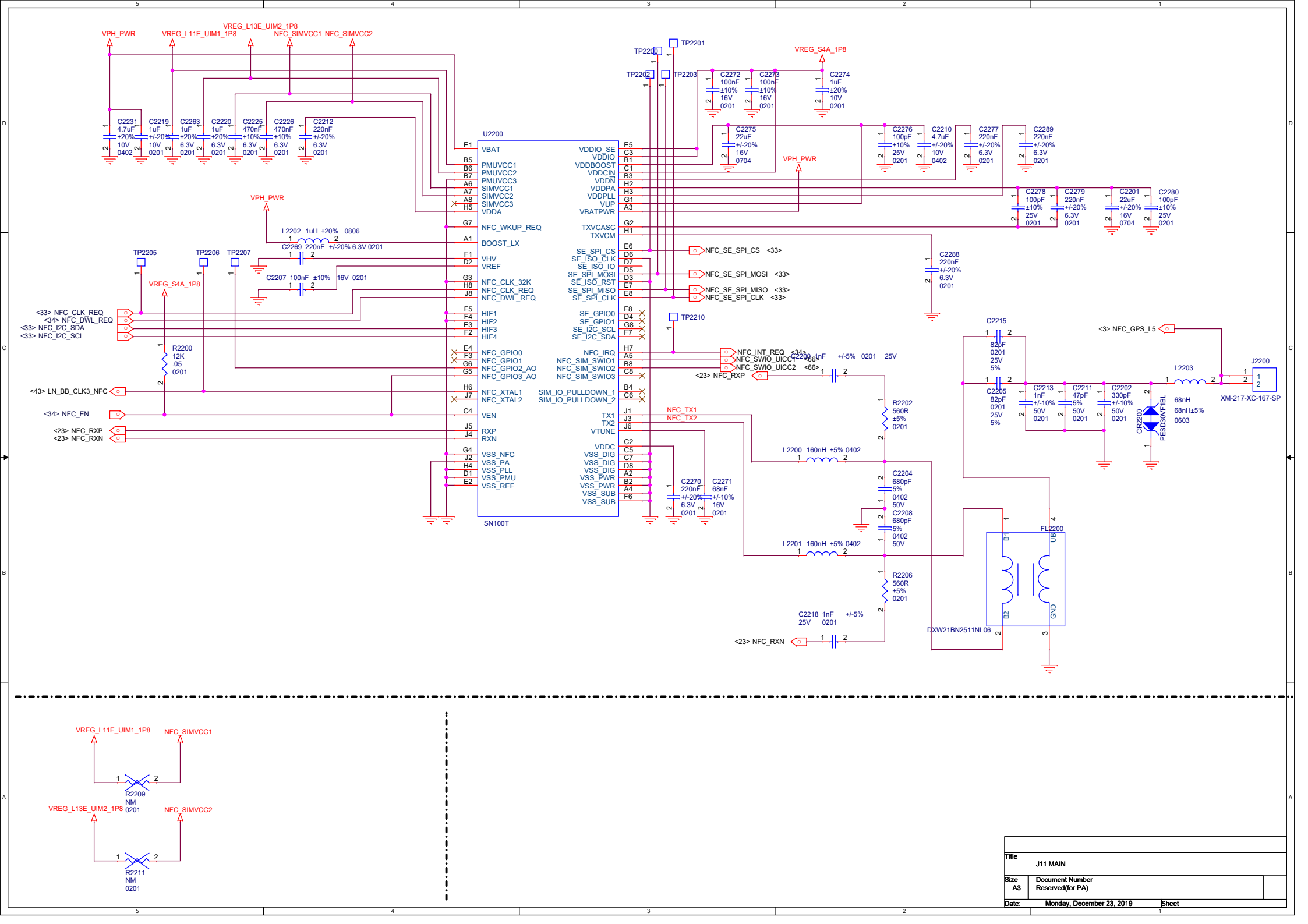


Title	
J11 MAIN	
Size	Document Number
A3	QLN DRX
Date: Monday, December 23, 2019	
Sheet	12 of 74

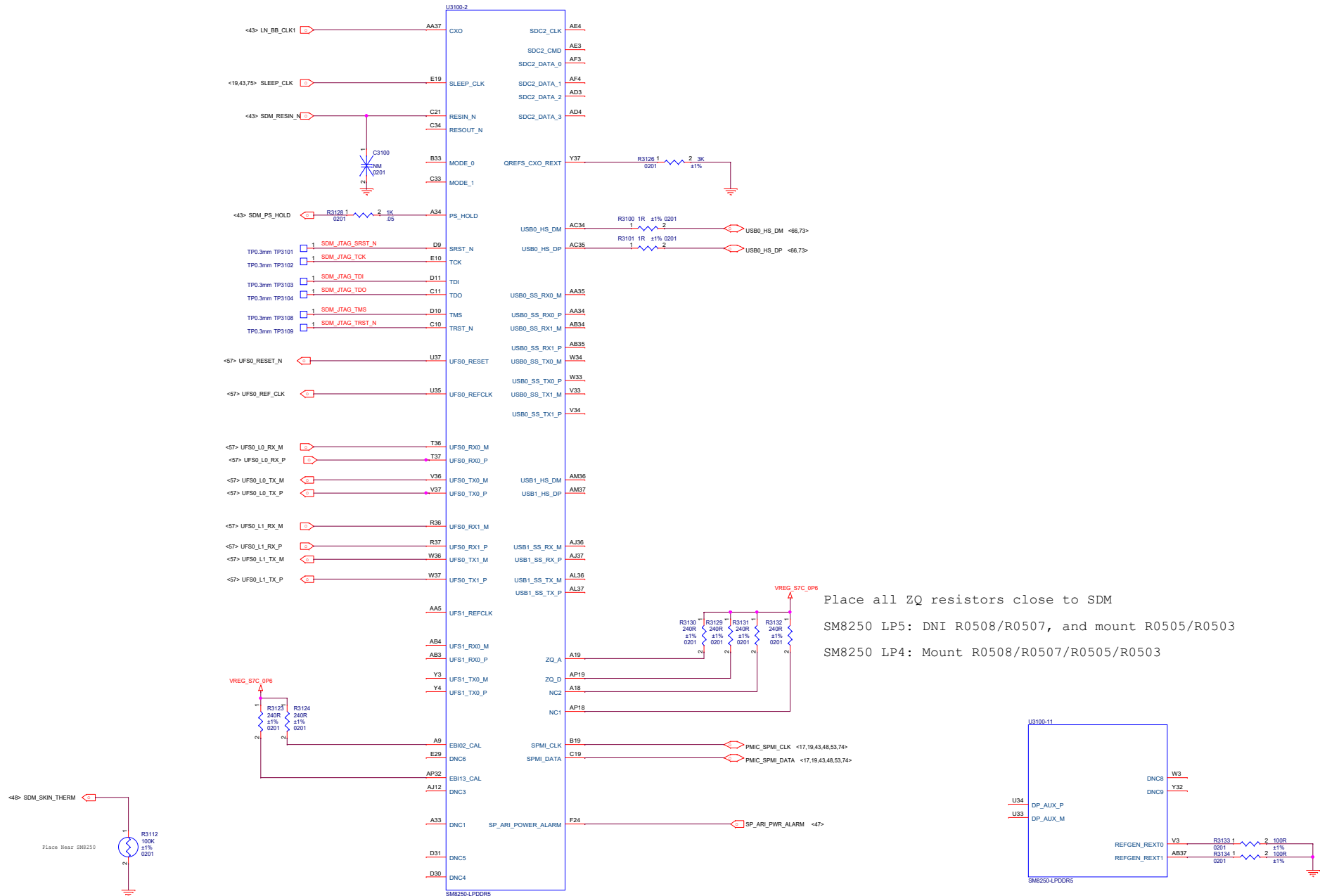


File	
J11 MAIN	
Size	Document Number
A2	Reserved for 5G-WTR P)
Date	Monday, December 23, 2019
Sheet	

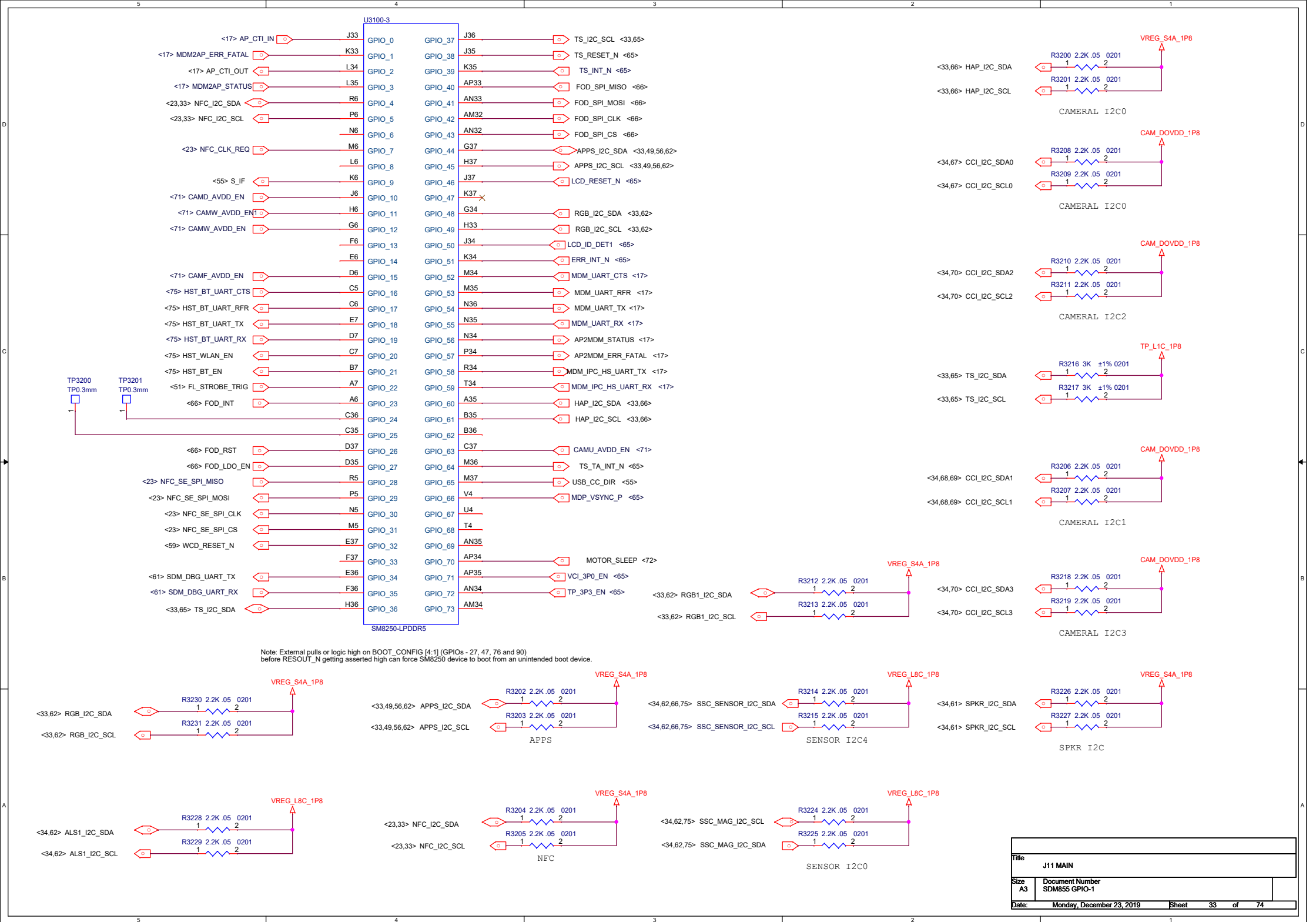




We use current SM8250 symbol as LP5 PKG, and it can be compatible with LP4x PKG



Title		J11 MAIN
Size	Document Number	SCHM855 Control
A2		
Date:	Monday, December 23, 2019	Sheet



REAR CAMERA WIDE
64M

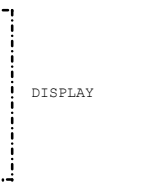
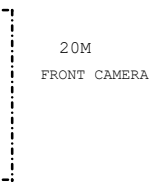
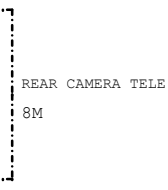
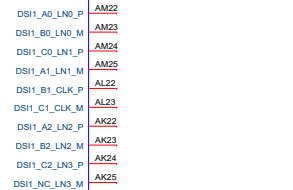
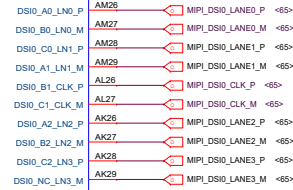
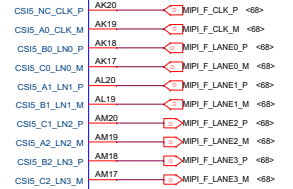
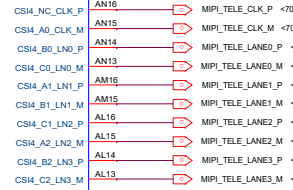
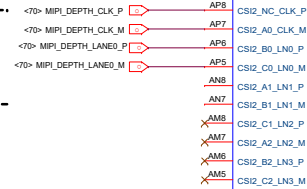
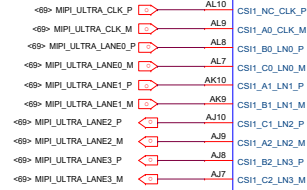
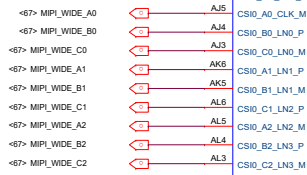
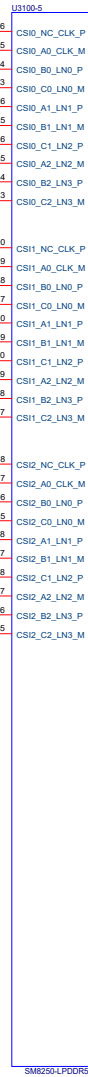
REAR CAMERA ULTRA 13M

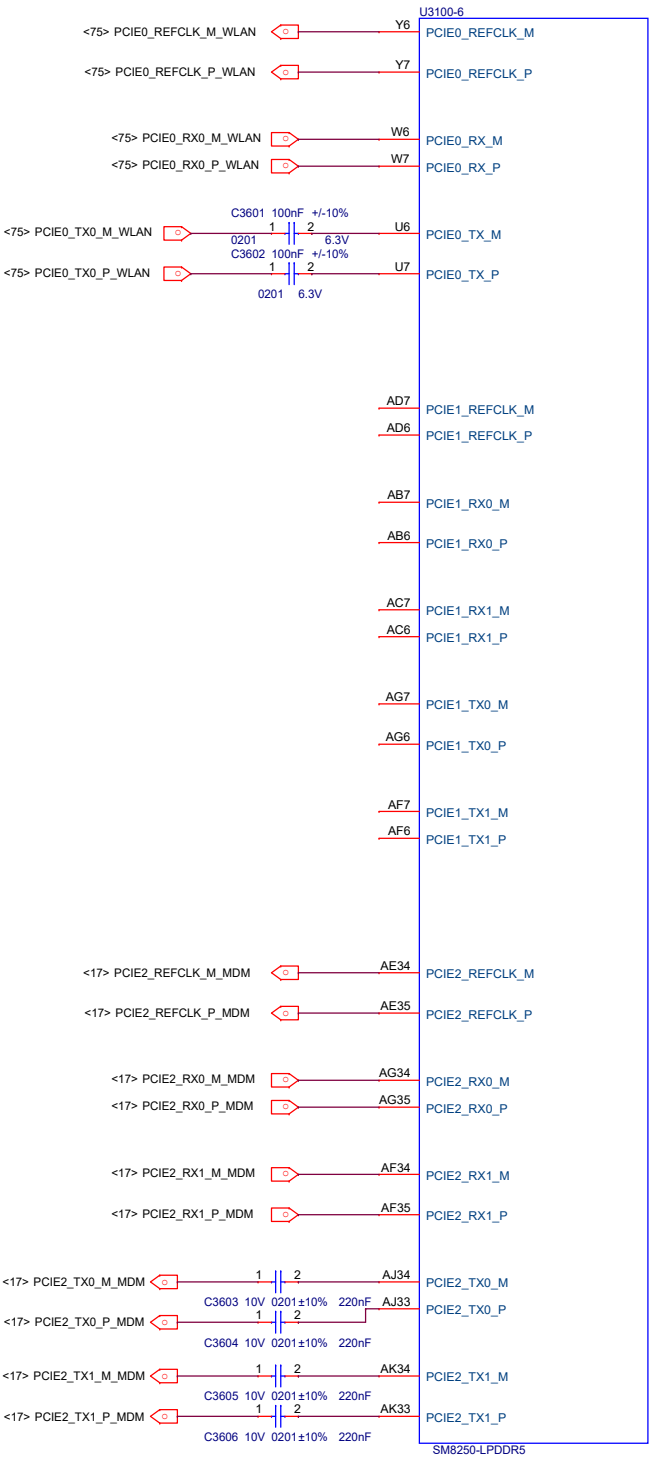
REAR CAMERA DEPTH
2M

REAR CAMERA TELE
8M

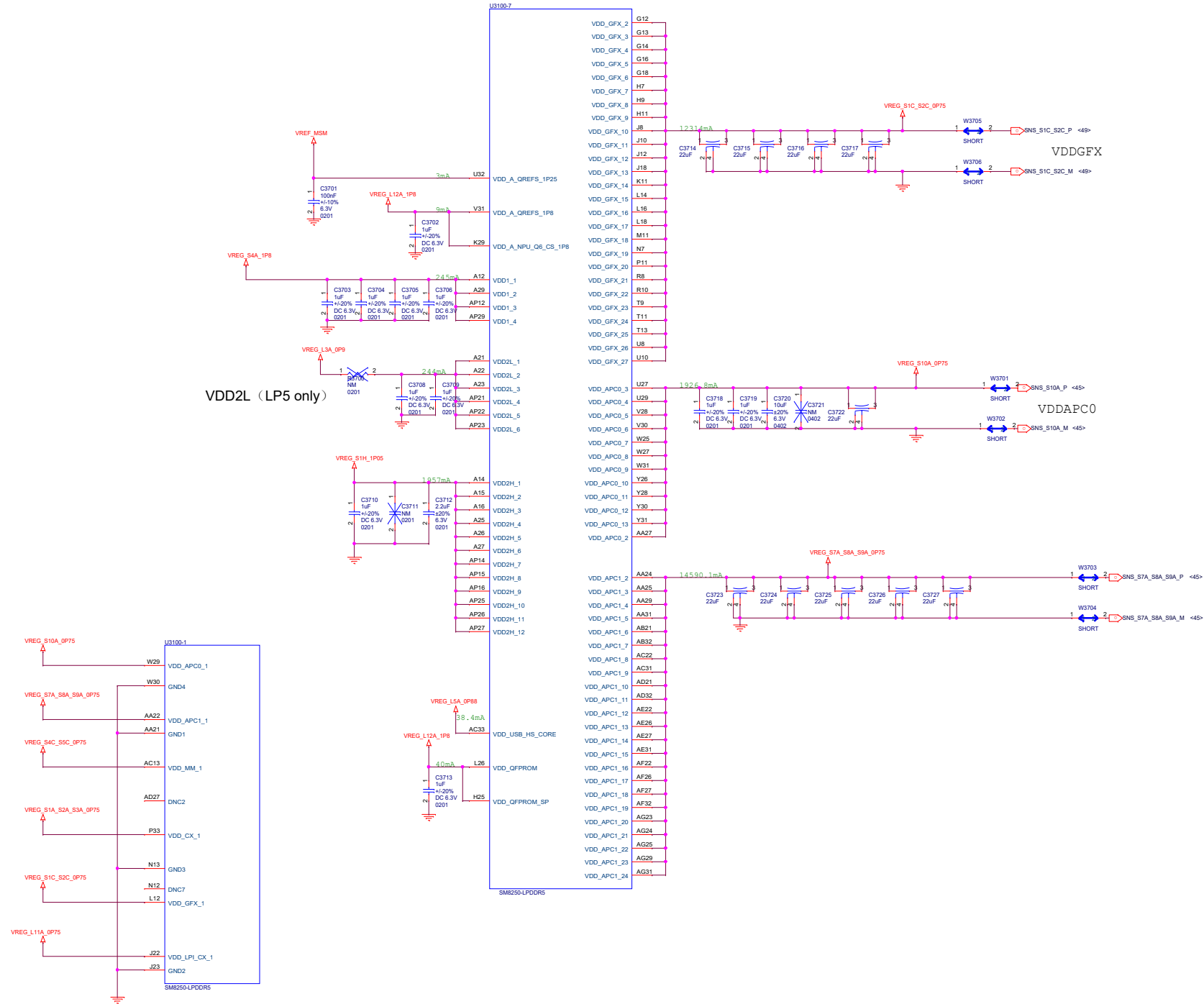
20M
FRONT CAMERA

DISPLAY



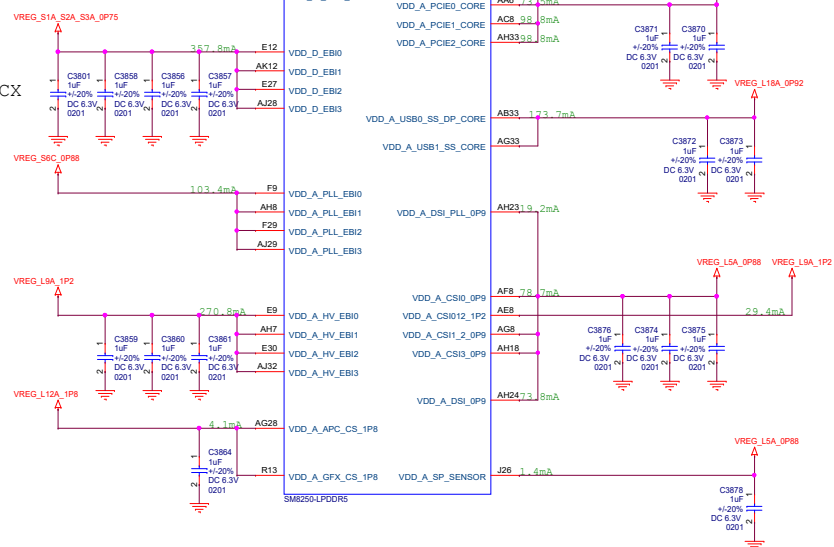
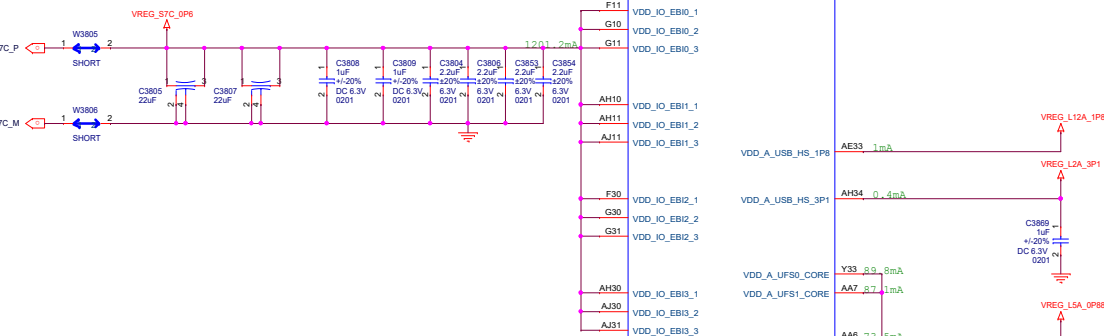
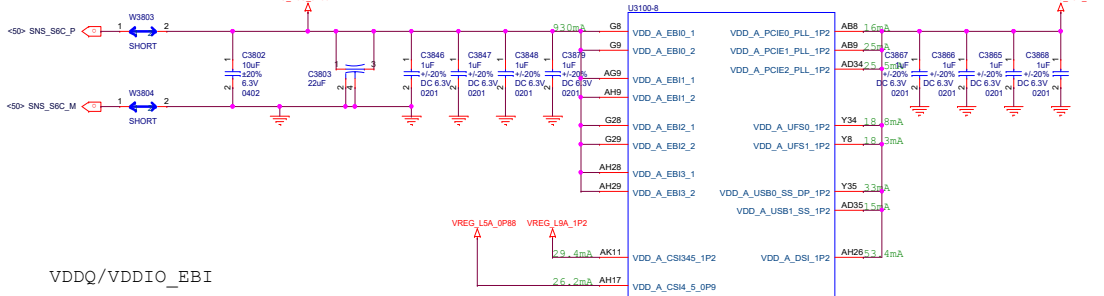


VDD2L (LP5 only)



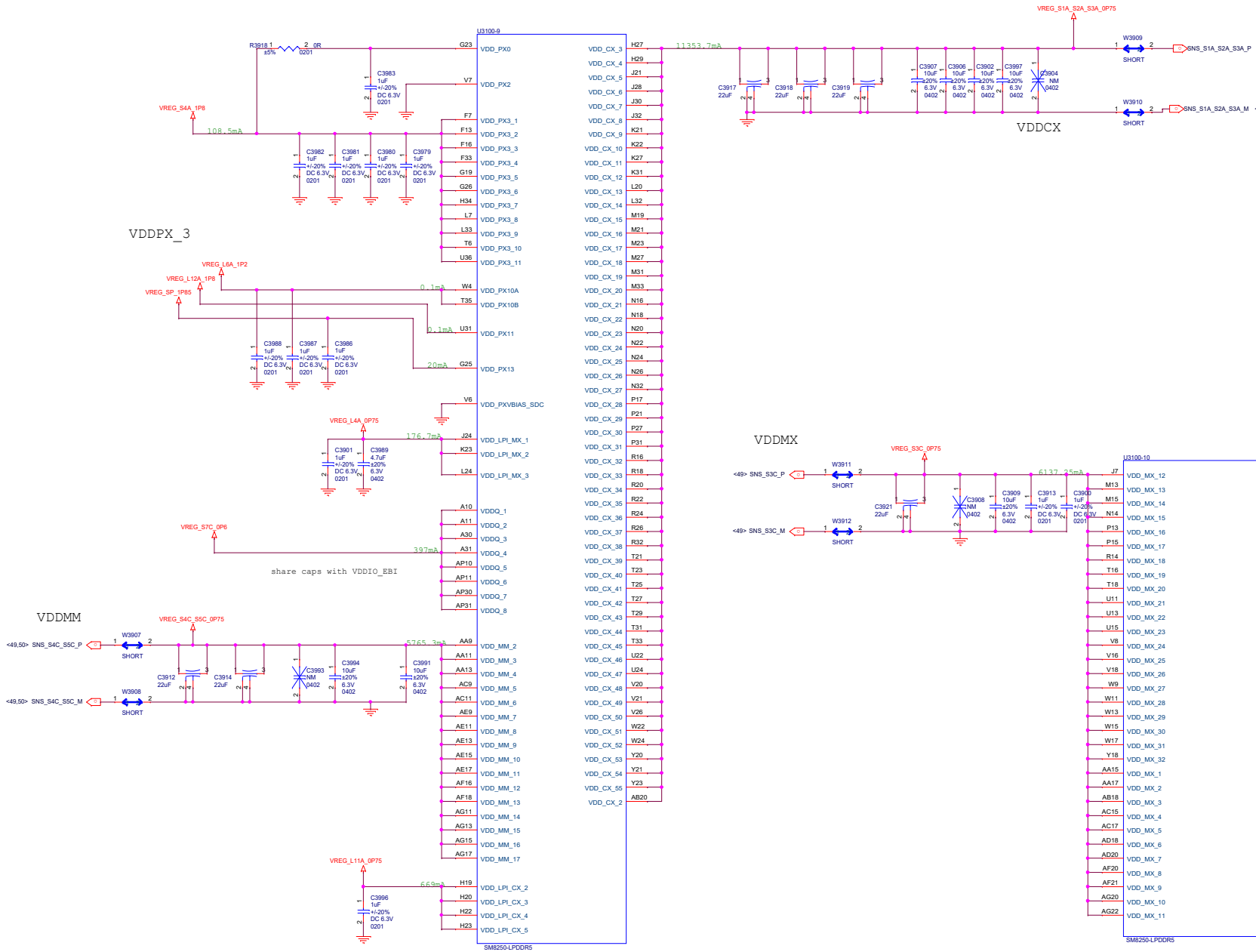
VDDAPC1

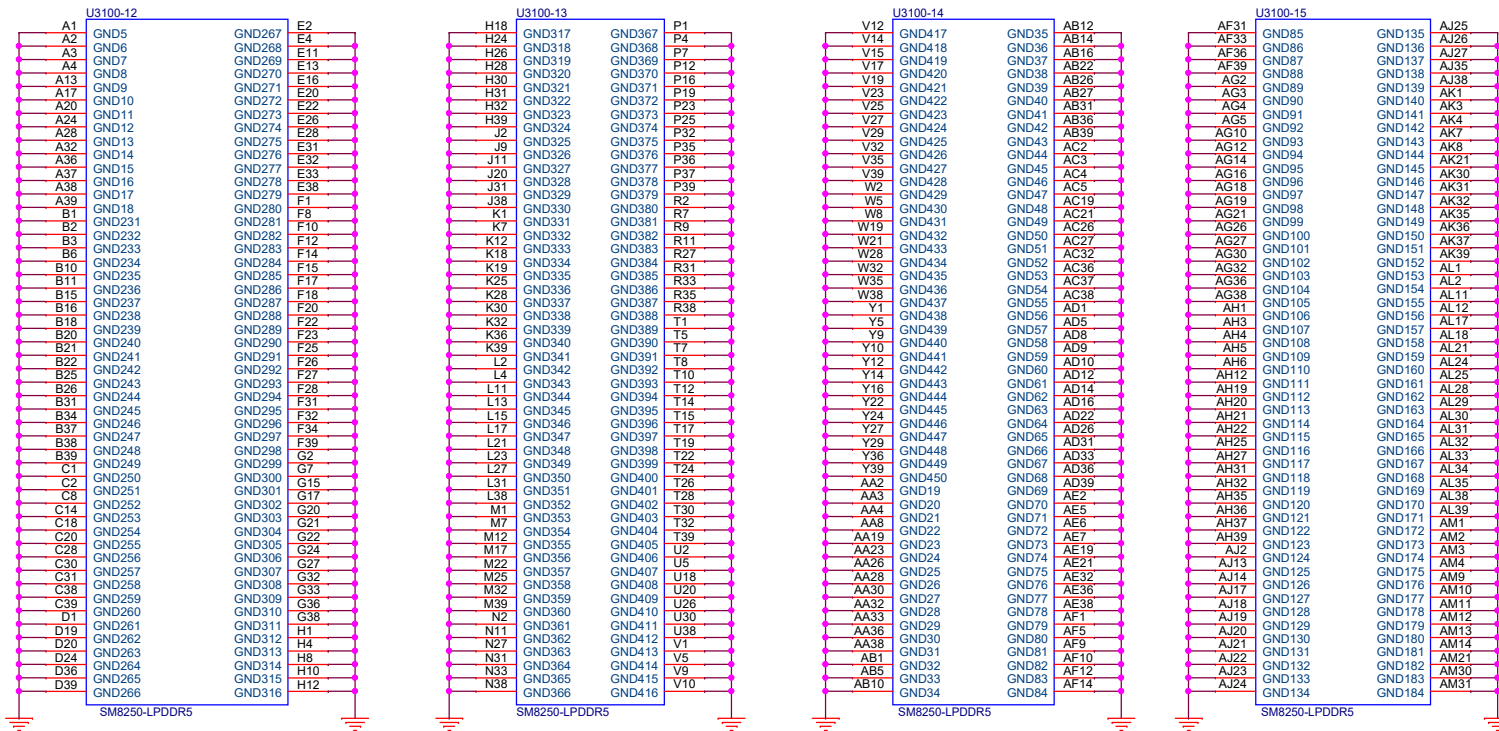
Title		J11 MAIN
Size	Document Number	SCHM55 PWR1
A2		
Date:	Monday, December 23, 2019	Sheet



Title		
J11 MAIN		
Size	Document Number	
A2	SDM855 PWR2	
Date:	Monday, December 23, 2019	Sheet

Title		J11 MAIN	
Size	Document Number		
A2	SDM855 PWR2		
Date:	Monday, December 23, 2019		Sheet

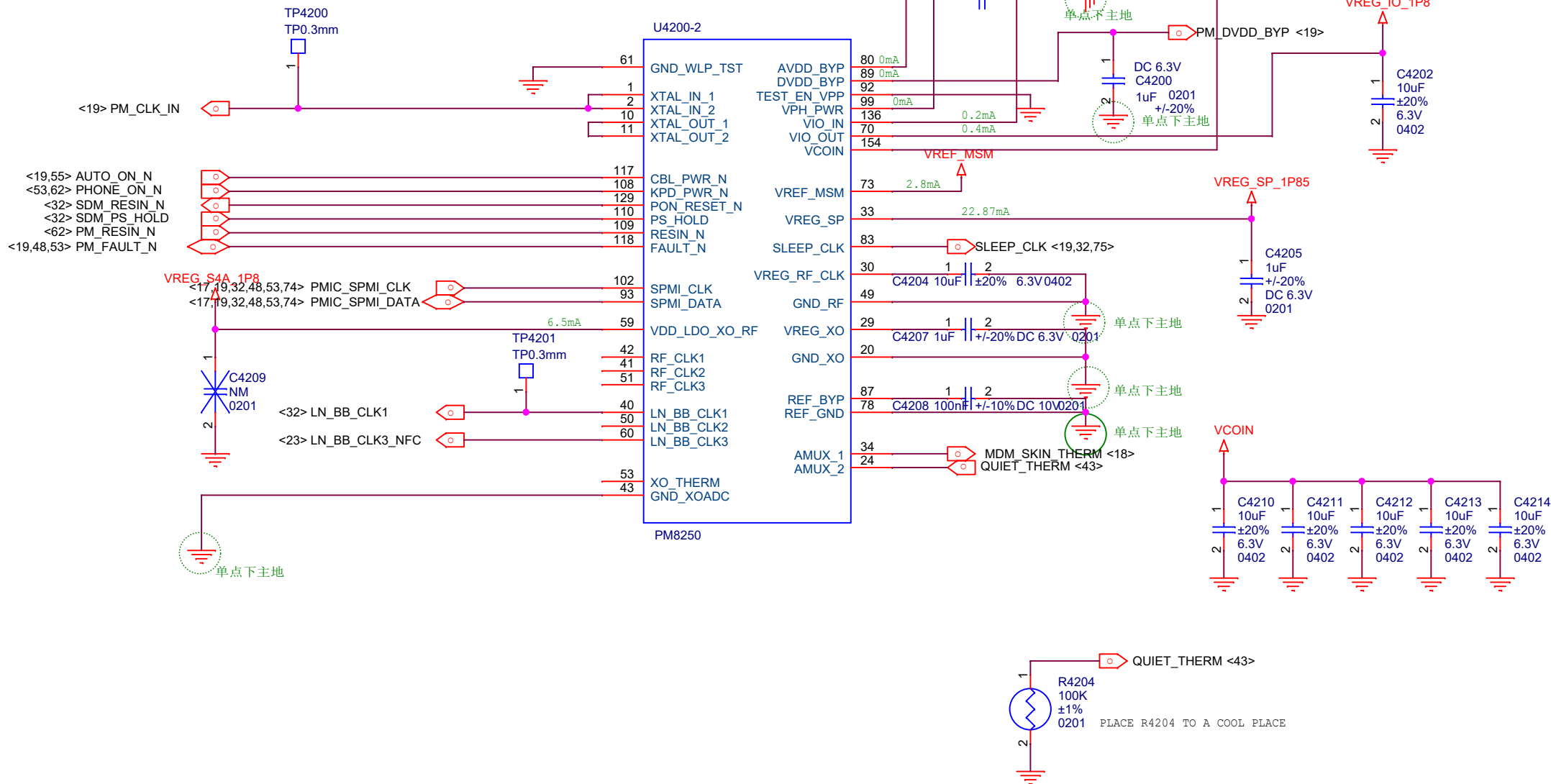




Note: Refer to 80-VP447-10 for XO Layout App Note.

Layout Note:

No trace shall be present in at-least two immediate layers below XTAL.
Route XTAL_IN and XTAL_OUT traces on surface layer only, with length
between 3mm and 10mm.



电容下主地

电容下主地

电容下主地

电容下主地

电容下主地

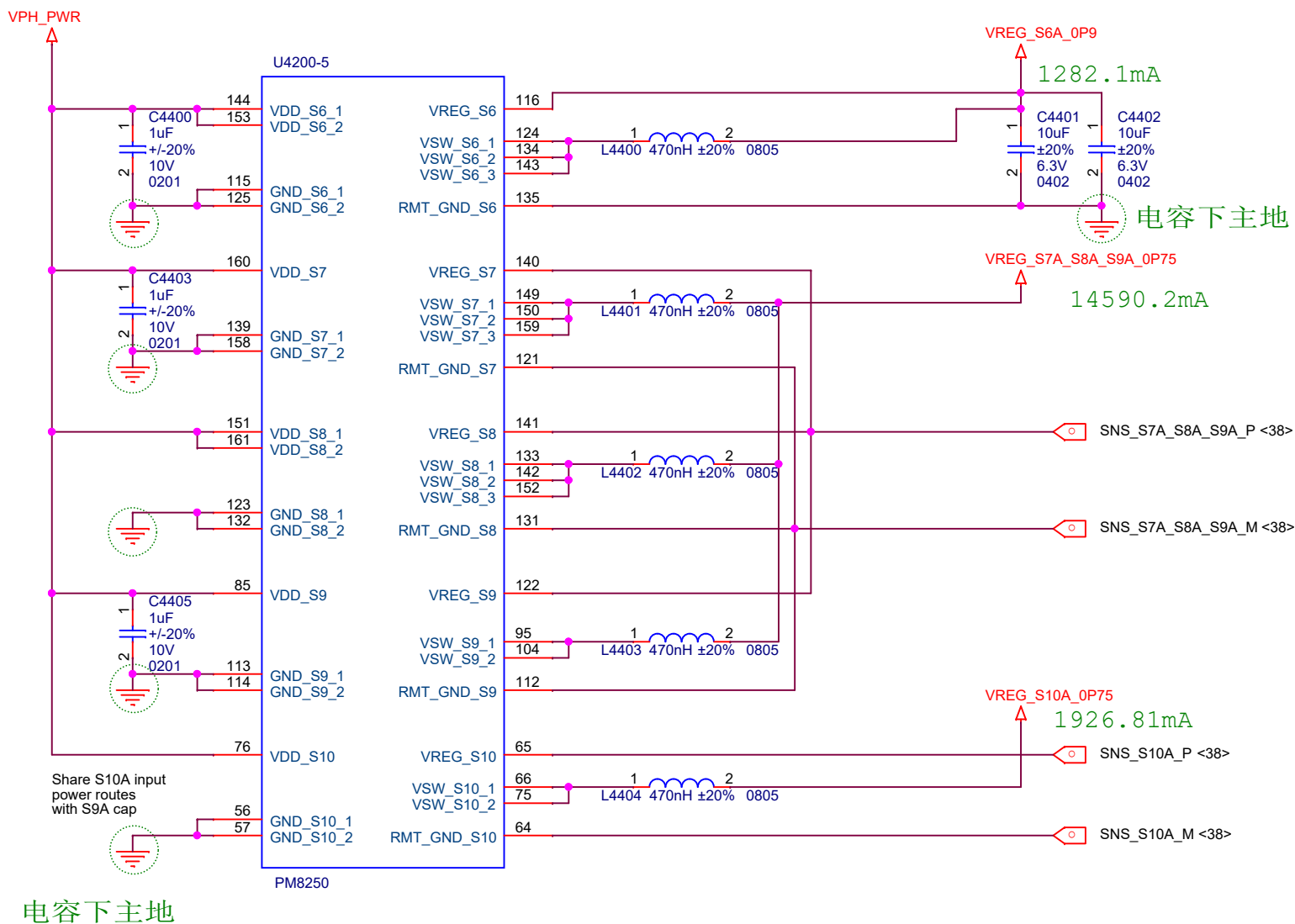
VREG_S1A_S2A_S3A_0P75
11711.5mA

VREG_S4A_1P8
2658.7mA

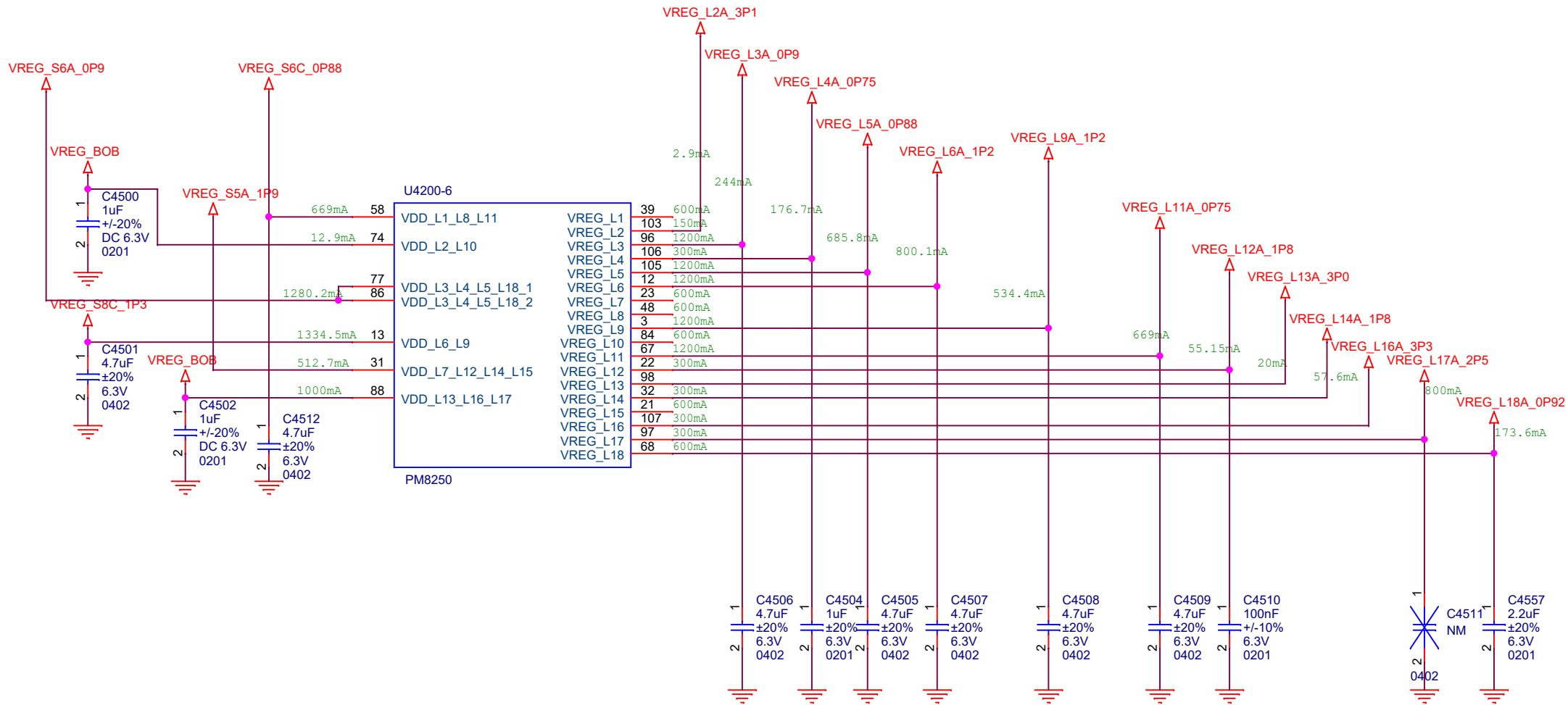
VREG_S5A_1P9
922.7mA

Title		
J11 MAIN		
Size	Document Number	
A4	PM855 BUCK CONVERTER1	
Date:	Monday, December 23, 2019	Sheet 44 of

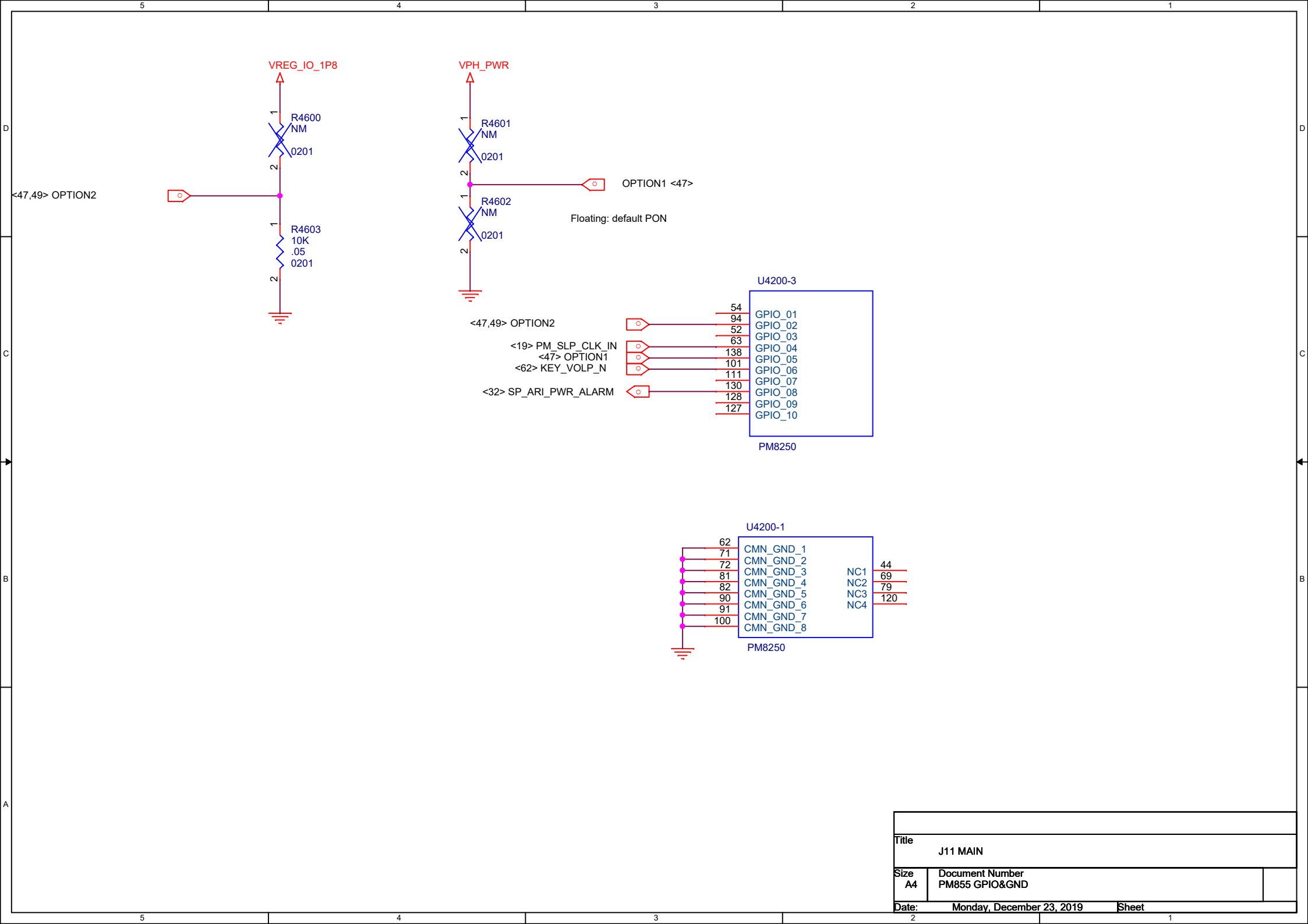
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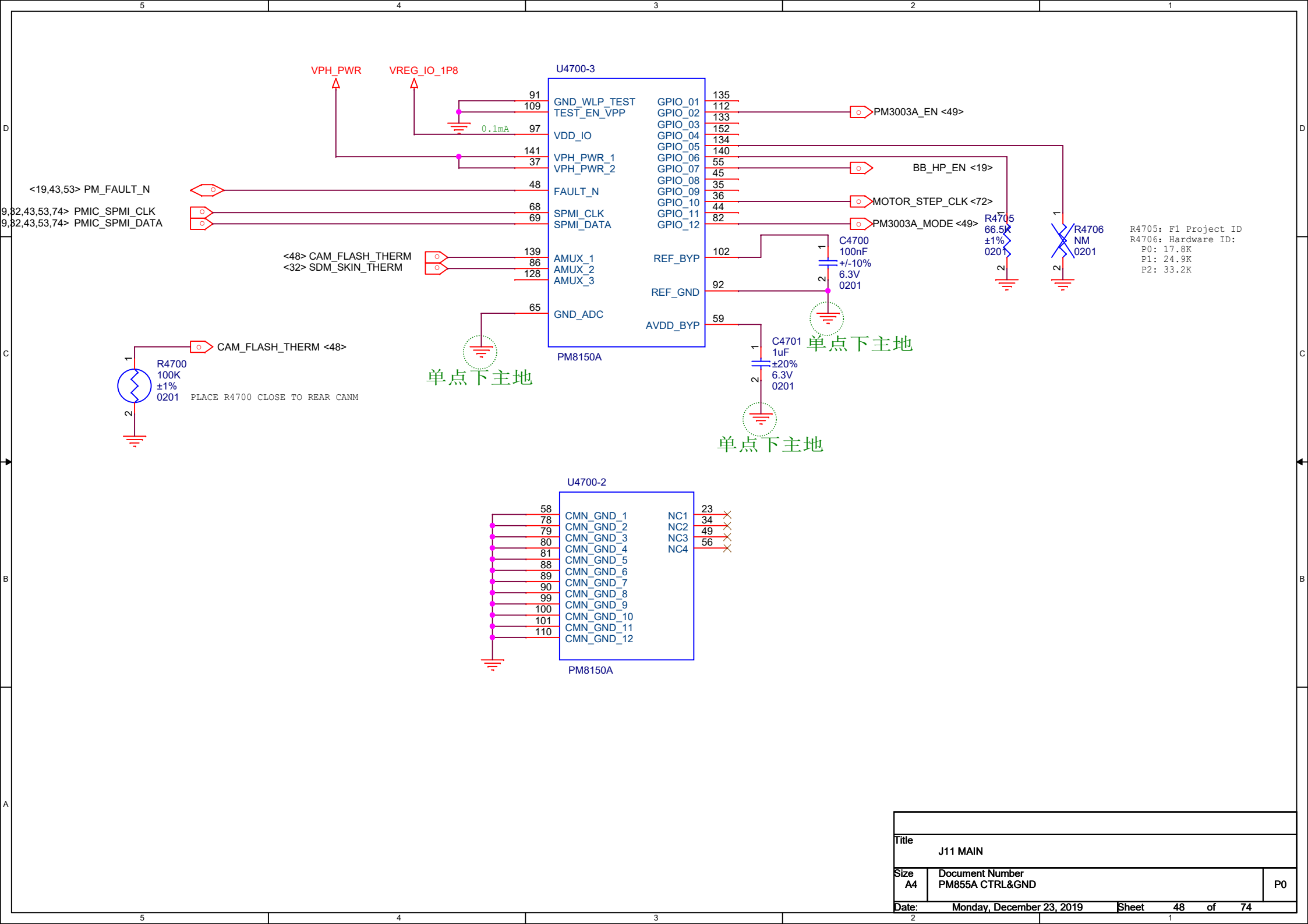


Title		
J11 MAIN		
Size	Document Number	P0
A4	PM855 BUCK CONVERTER2	
Date:	Monday, December 23, 2019	Sheet 45 of 74



Pseudo capless allows users to place the LDO output capacitor physically/electrically far from the LDO output, near the load. The routing ESL/ESR between the PMIC pin and the remote Cout should meet the capless LDO spec. All PMOS LDOs are considered pseudo-capless, unless it has internal PMIC load. If an internal PMIC load, the output cap must be placed close to the PMIC. Required capacitance for each LDO type can be found in App Note: 80-VT310-125





VPH_PWR

C4806
10uF
+/-20%
10V
0402

电容下主地

Note: Similar to PM8250, please ensure that a minimum of three 10uF bulk capacitors are present on PM8150L/A VPH_PWR primarily to support the SMPS buck regulators even if BoB or Lighting Modules are unused. For PM8150L, the three 10uF bulk capacitors on VPH_PWR are fulfilled by the 10uF capacitors for BoB, WLED, and LCDB. For the PM8150A, the three 10uF bulk capacitors on VPH_PWR are fulfilled by the 10uF caps for BoB, OLEDB, ELVDD, and ELVSS.

电容下主地

U4700-5

VDD_S1_1
VDD_S1_2GND_S1_1
GND_S1_2VDD_S2_1
VDD_S2_2
VDD_S2_3GND_S2_1
GND_S2_2

VDD_S3_1

GND_S3_1
GND_S3_2VDD_S4_1
VDD_S4_2

PM8150A

VREG_S1

VSW_S1_1
VSW_S1_2

RMT_GND_S1

VREG_S2

VSW_S2_1
VSW_S2_2
VSW_S2_3

RMT_GND_S2

VREG_S3

VSW_S3_1
VSW_S3_2

RMT_GND_S3

VREG_S4

VSW_S4_1
VSW_S4_2

RMT_GND_S4

SNS_S1C_S2C_P <38>

12314mA

SNS_S1C_S2C_M <38>

VREG_S3C_0P75

6137.4mA

SNS_S3C_P <40>

SNS_S3C_M <40>

VREG_S4C_S5C_0P75

5165.3mA

SNS_S4C_S5C_P <40,50>

SNS_S4C_S5C_M <40,50>

VPH_PWR

C4801
10uF
0402
10V
±20%C4803
100nF
+/-10%
DC 10V
0201<33,56,62> APPS_I2C_SDA
<33,56,62> APPS_I2C_SCLR4802 NM 0201
R4803 NM 0201R4800 0R
±5%
0201
R4801 0R
±5%
0201

U4800

PVIN
PVIN
PVIN
VSEL
EN
SDA
SCL
AGND
PGND
PGND
PGND

PM3003A

SW1
SW2

MODE

VOUT

AGND
PGND
PGND
PGND

下主地

下主地

L4804
470nH
±20%
0805C4804
22uFC4805
22uF

C4807 NM 0402

C4808 NM 0402

VREG_S1H_1P05

1957mA

VREG_S1H_1P05

PM3003A_MODE <48>

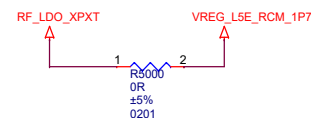
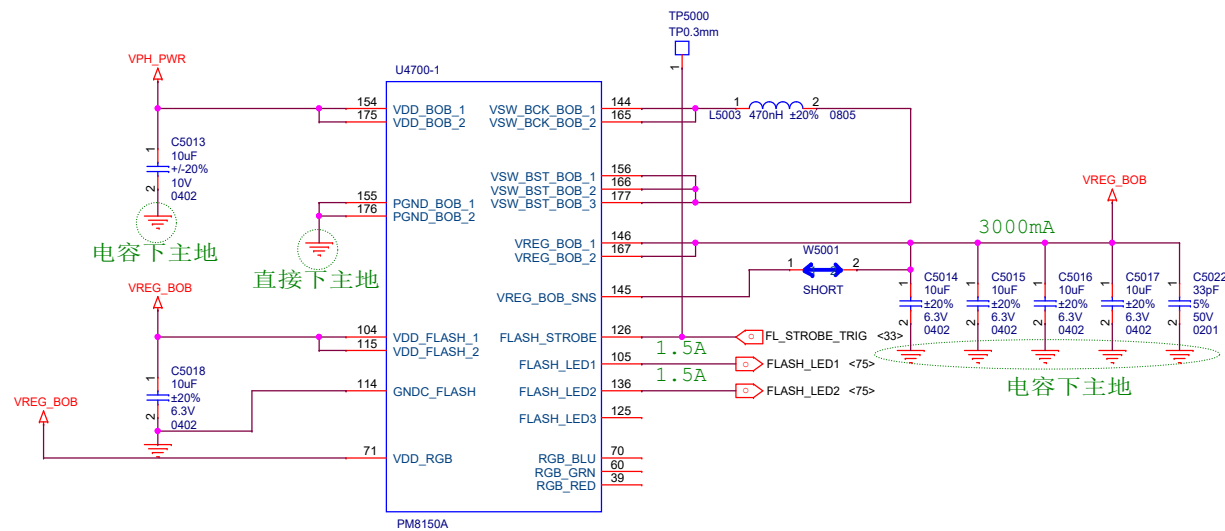
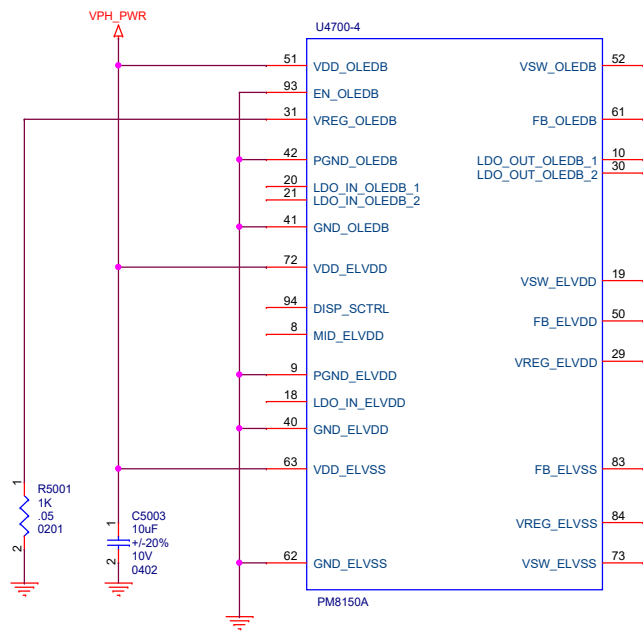
Title		J11 MAIN	
Size	A4	Document Number	PM855A BUCK CONVERTER1
Date:	Monday, December 23, 2019		Sheet 49 of 74

电容下主地

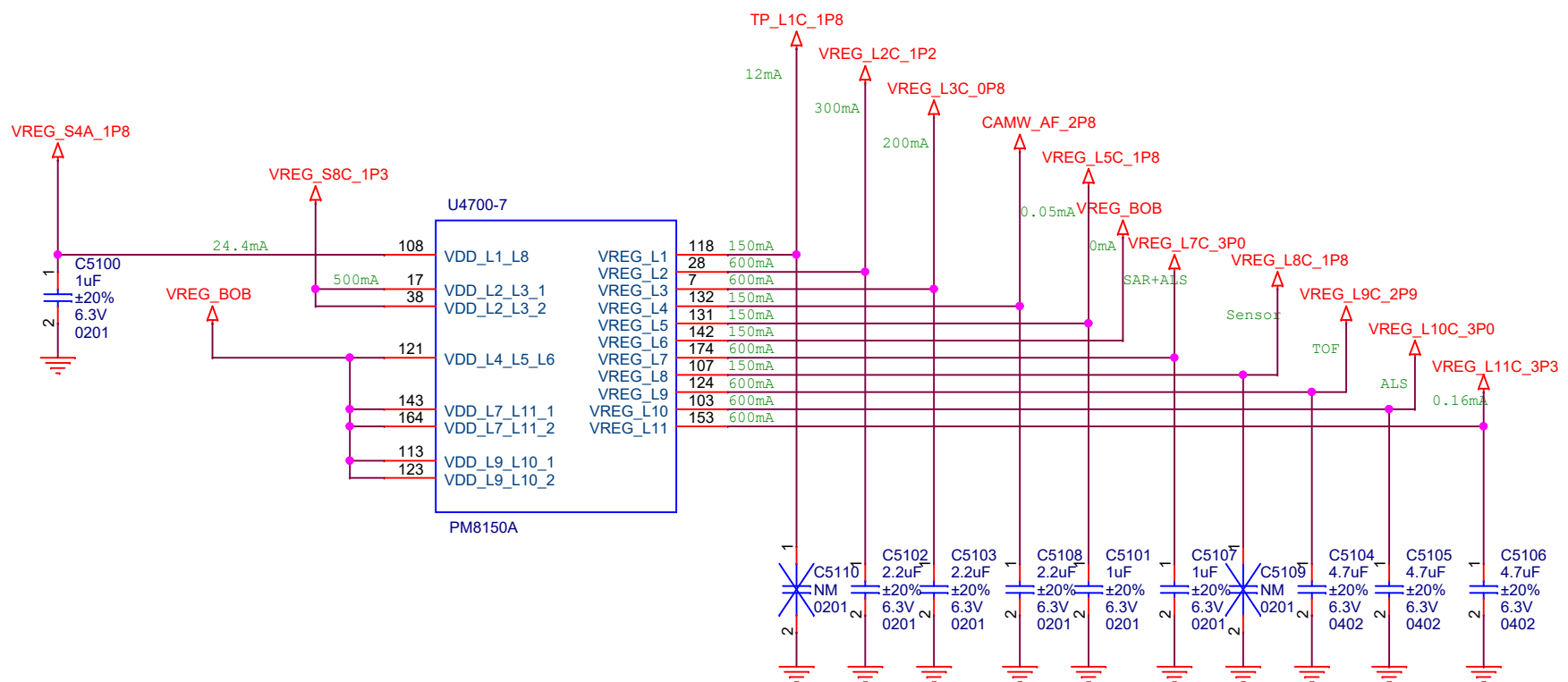
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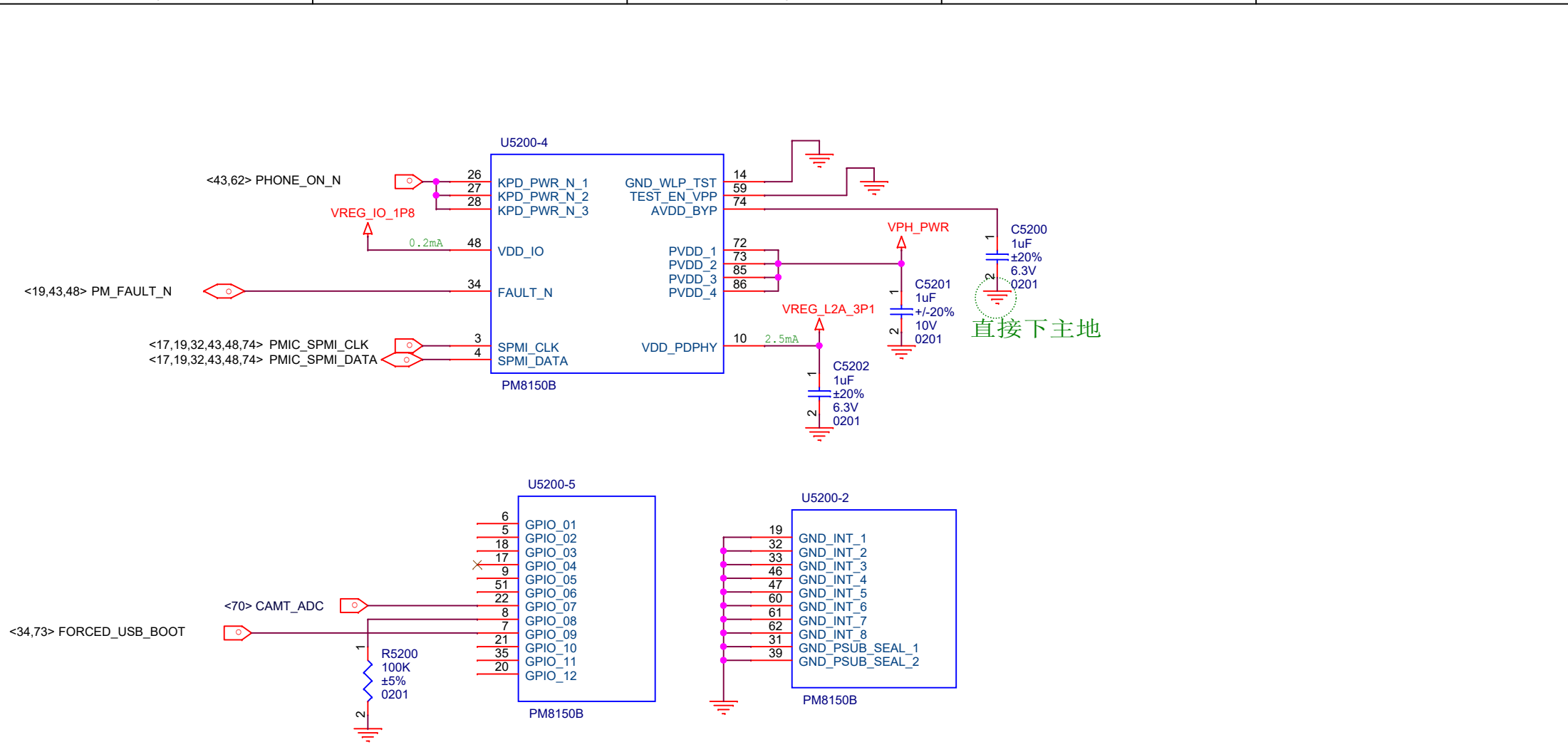
电容下主地

Title		
J11 MAIN		
Size	Document Number	P0
A4	PM855A BUCK CONVERTER2	
Date:	Monday, December 23, 2019	Sheet 50 of 74

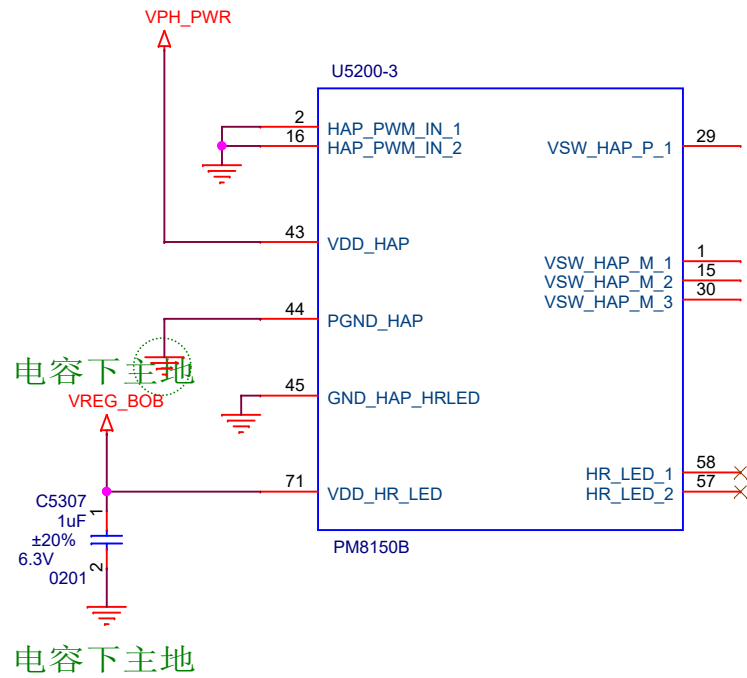


Title	
J11 MAIN	
Size	Document Number
A3	PM855A AMOLED&BOB&FLASH
Date:	Monday, December 23, 2019
Sheet	51 of 74

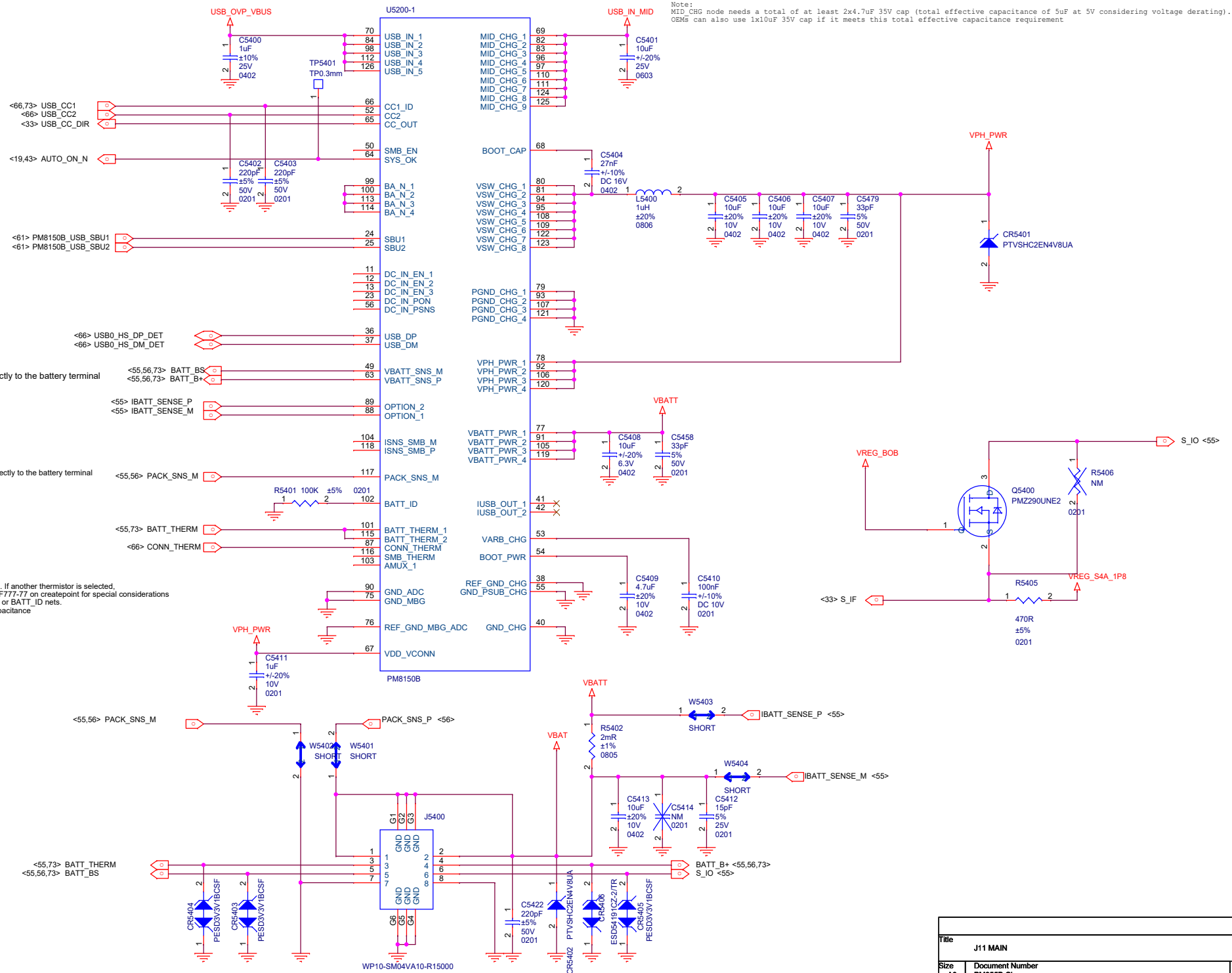




Note: GPIO_04 and GPIO_11 of PM8150B are for internal usage. Keep them float.
Keep GPIO_03 of PM8150B float. Don't use it for other purpose.
GPIO_06 of PM8150B is dedicated for SMB_STAT. Don't use it for other purpose.



Title			
J11 MAIN			
Size	Document Number		
A4	PM855B HAPTIC		
Date:	Monday, December 23, 2019	Sheet	54 of 74

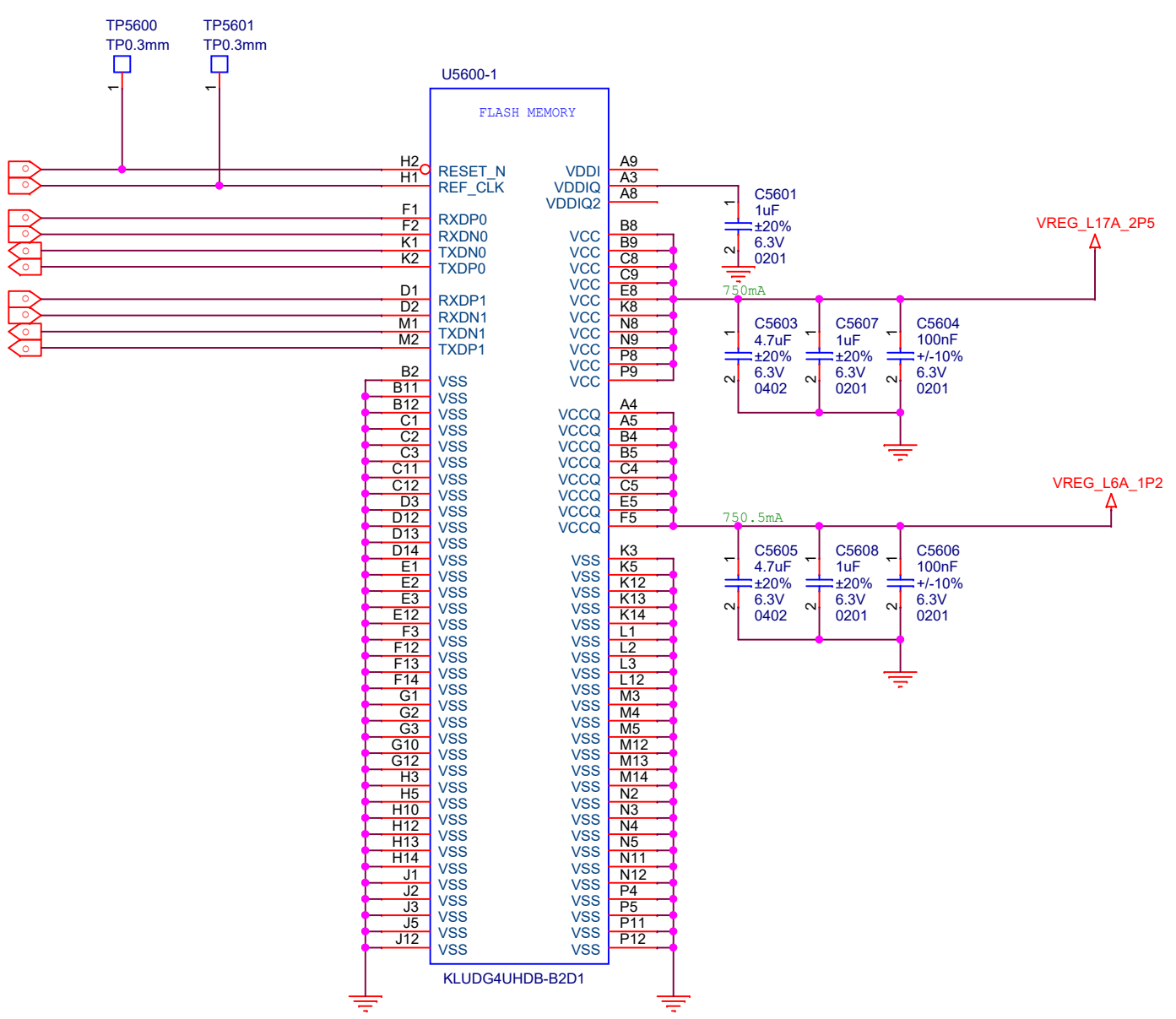


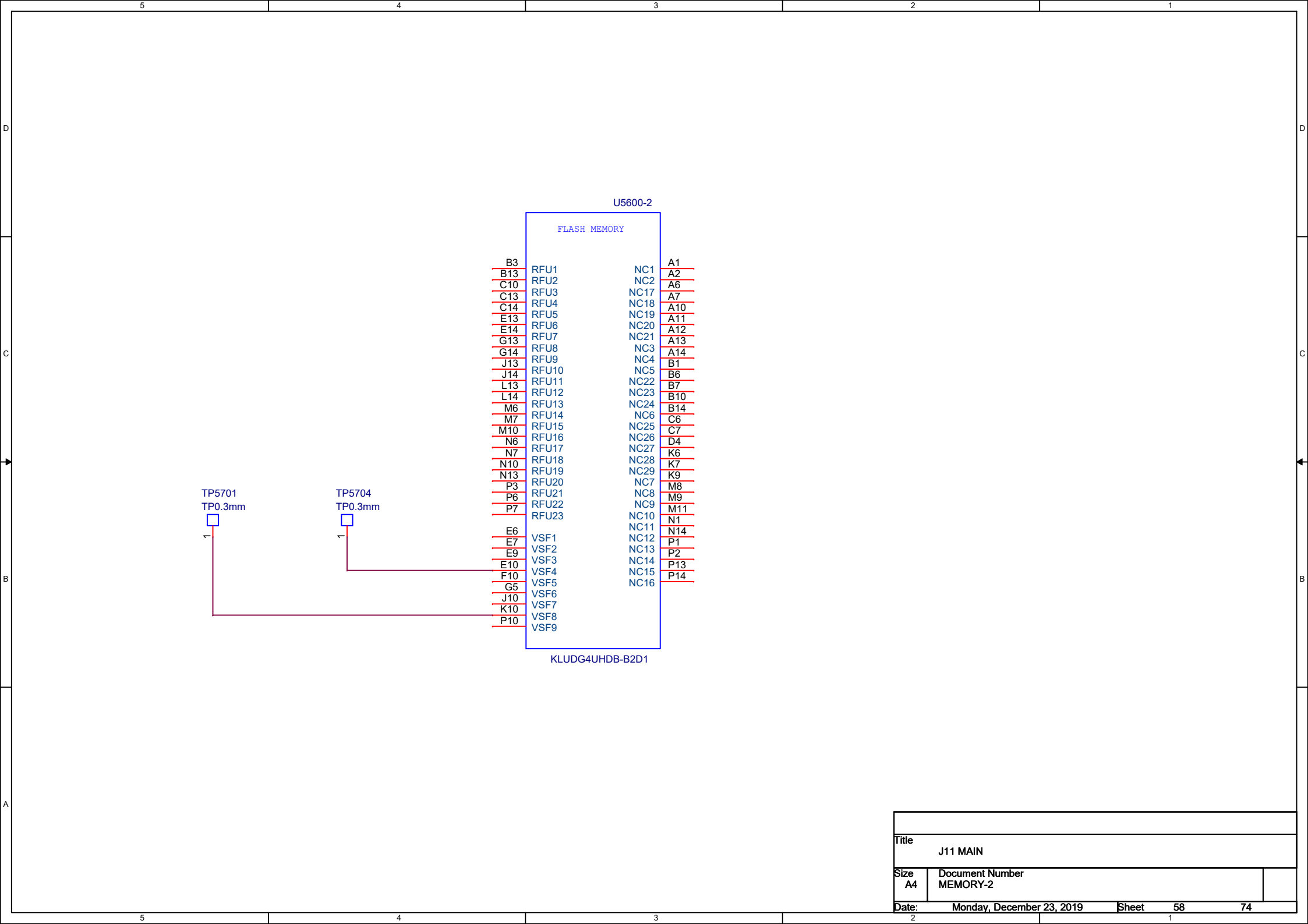
Title		J11 MAIN
Size		A3
Document Number		PM855B Charger
Date:		Monday, December 23, 2019
Sheet		1

<32> UFS0_RESET_N
<32> UFS0_REF_CLK

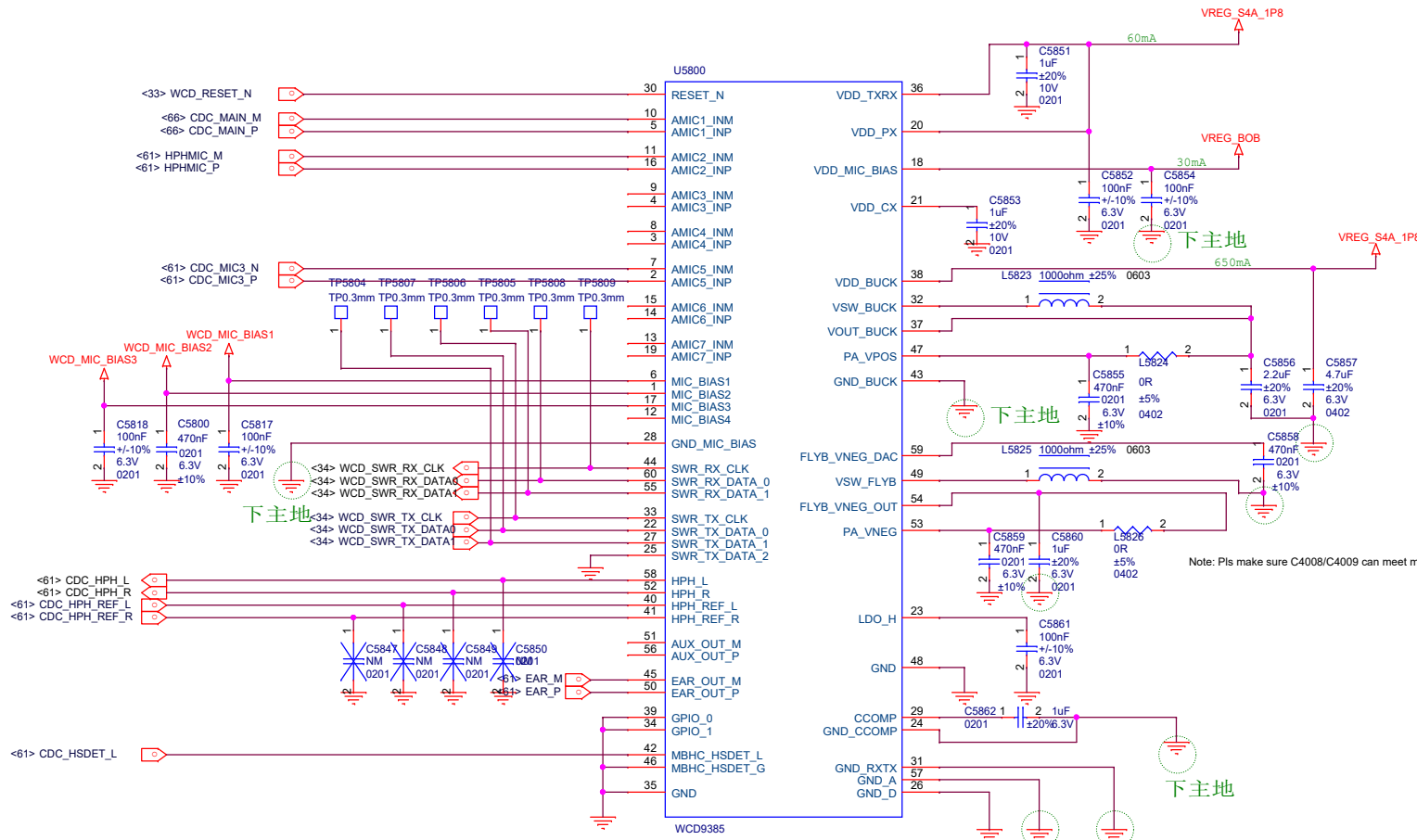
<32> UFS0_L0_TX_P
<32> UFS0_L0_TX_M
<32> UFS0_L0_RX_M
<32> UFS0_L0_RX_P

<32> UFS0_L1_TX_P
<32> UFS0_L1_TX_M
<32> UFS0_L1_RX_M
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Title			
J11 MAIN			
Size	Document Number		
A4	MEMORY-2		
Date:	Monday, December 23, 2019	Sheet	58 74

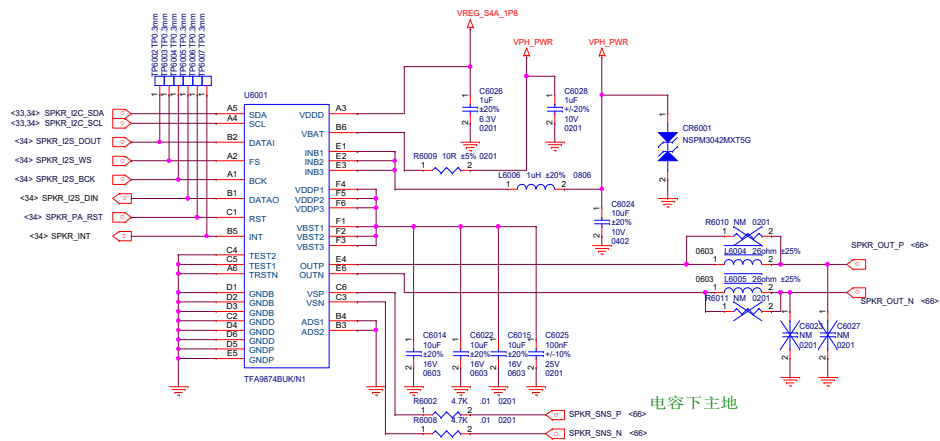


NOTE 1: ROUTE AN ISOLATED GROUND TRACE FROM CCOMP_REF TO THE CAP GROUND, THEN VIA TO THE MAIN GROUND PLANE.
NOTE 2: PLACE VDD_BUCK CAP AS CLOSE TO CHIP EDGE AS POSSIBLE AND MINIMIZE LOOP INDUCTANCE
NOTE 3: USE SHORT AND WIDE TRACES FOR BUCK AND FLYBACK RELATED NETS
NOTE 4: ANALOG MIC INPUTS SHOULD BE ISOLATED WITH GROUND FROM NOISY OR SWITCHING SIGNALS AND SUPPLIES

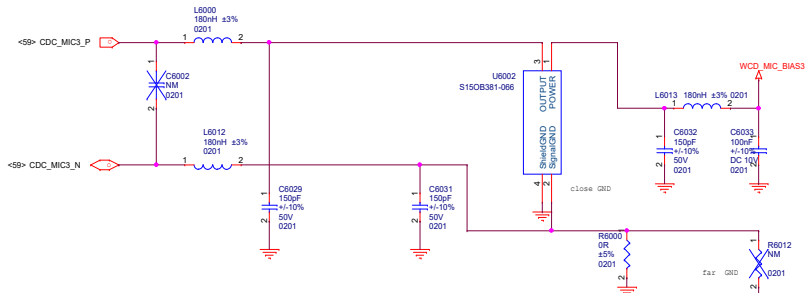
Note: Pls make sure C4008/C4009 can meet minimal capacitance requirement (0.4uF)@1.8V whether your use 1uF or 2.2uF

Title		J11 MAIN
Size	A3	Document Number
Date:		Monday, December 23, 2019
Sheet		1

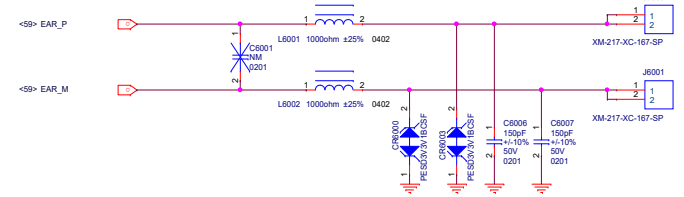
SPKR PA



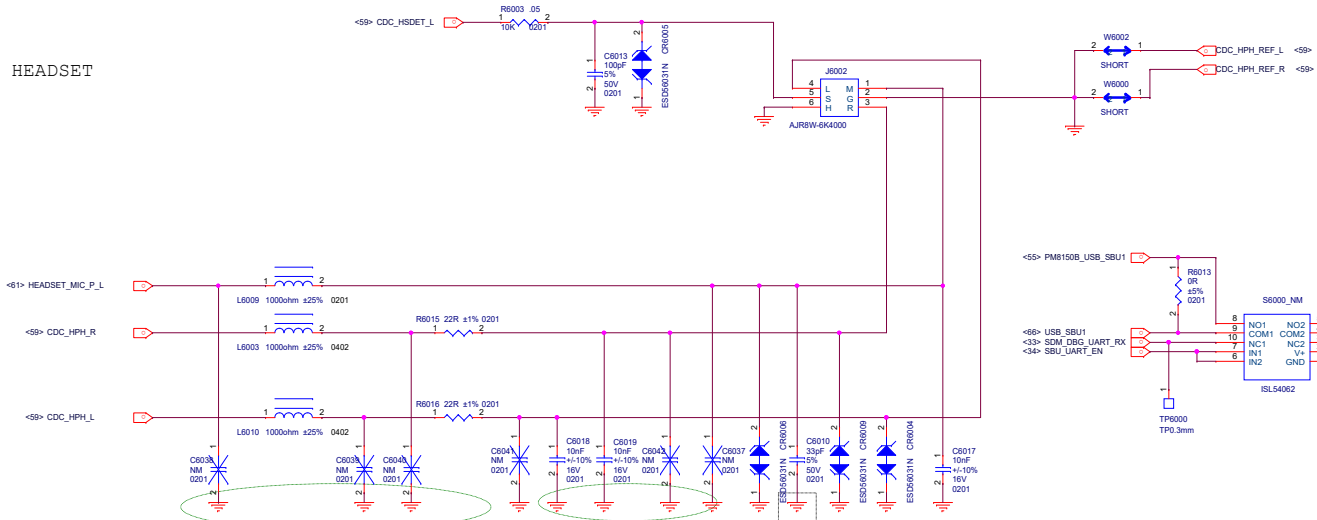
AUX MIC



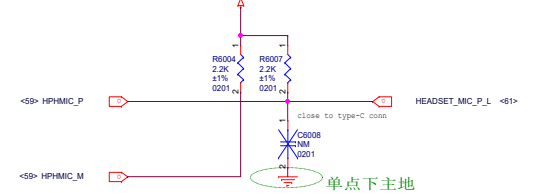
Receiver



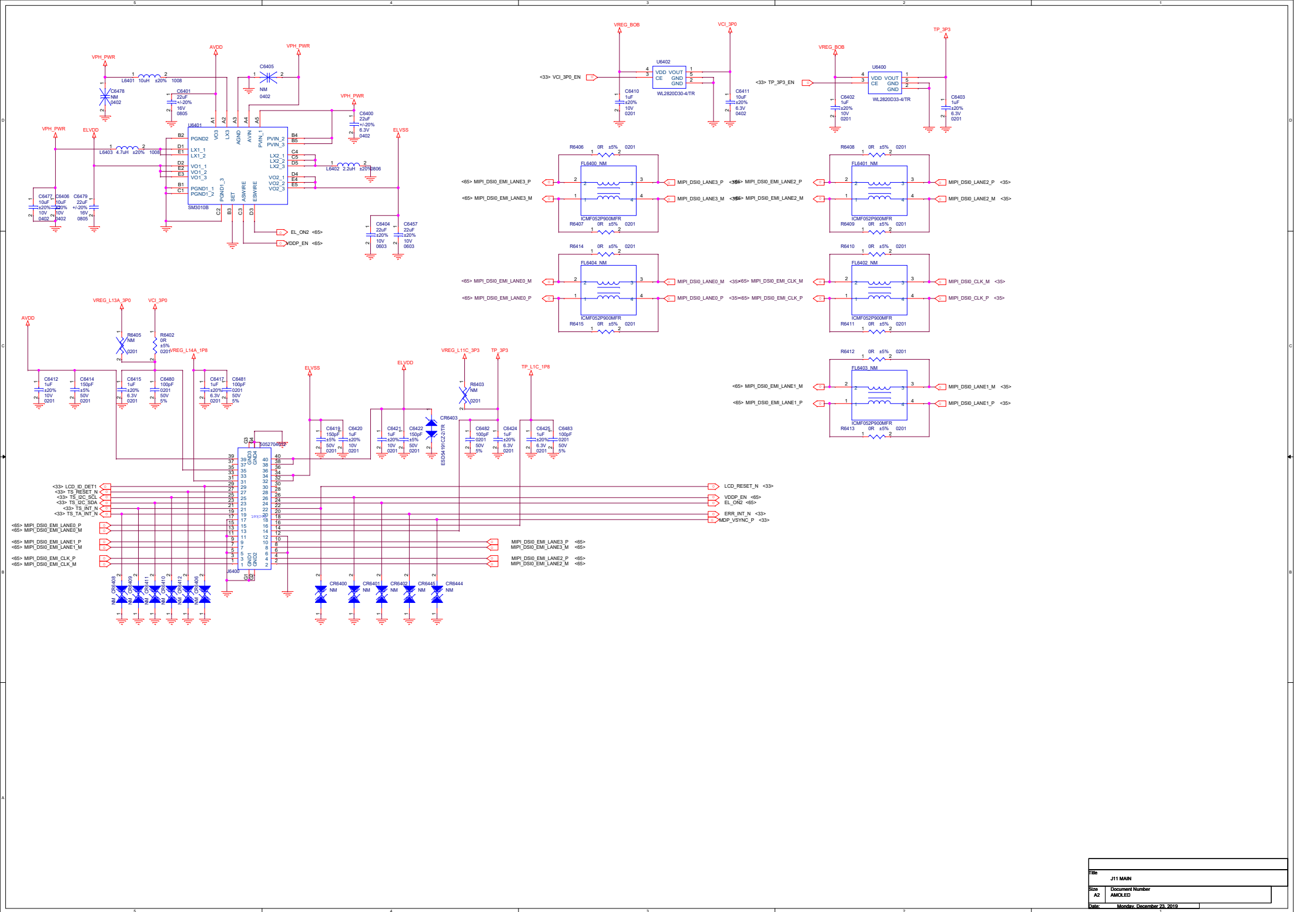
HEADSET



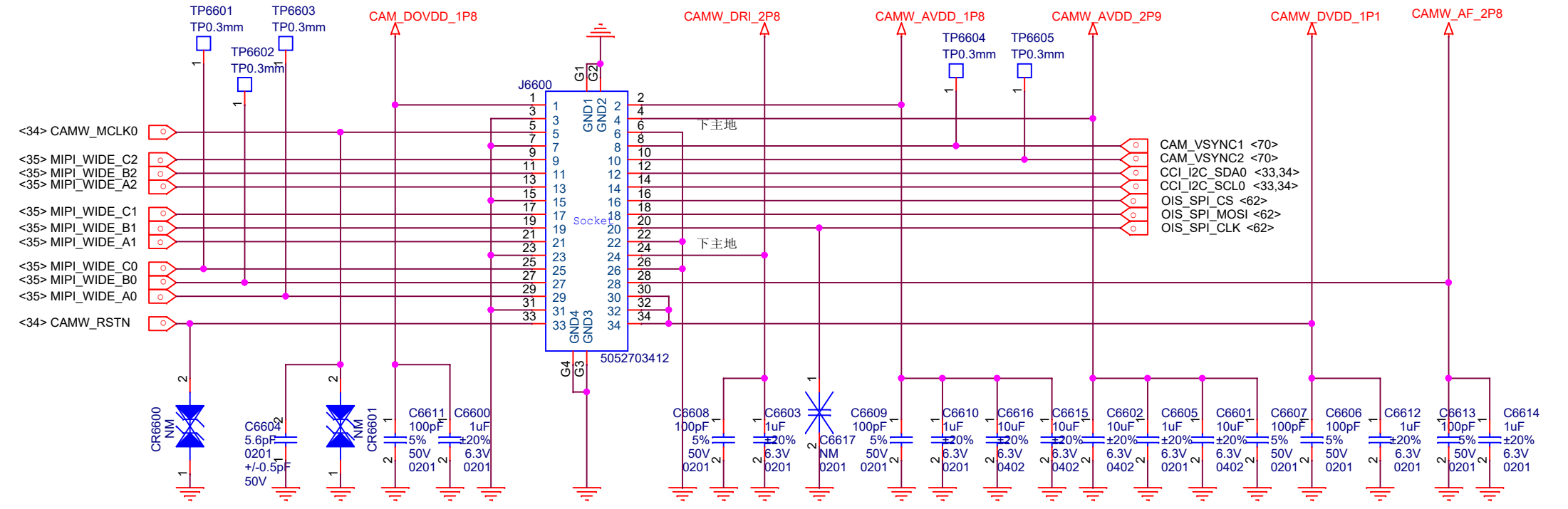
HEADSET



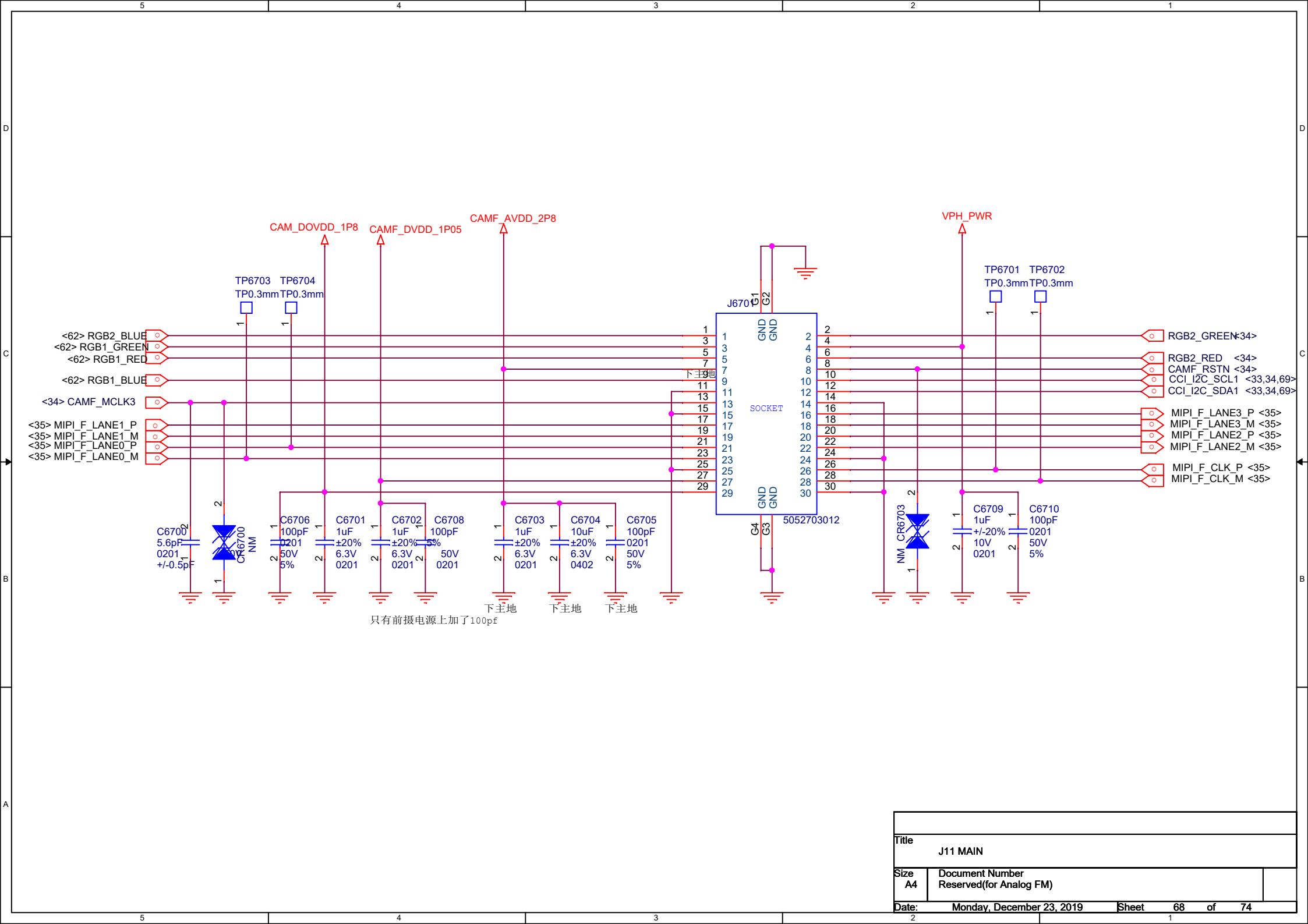
File	J11 MAIN
Size	Document Number
A2	Reserved (for AUDIO)
Date	Monday, December 23, 2019
Sheet	Sheet



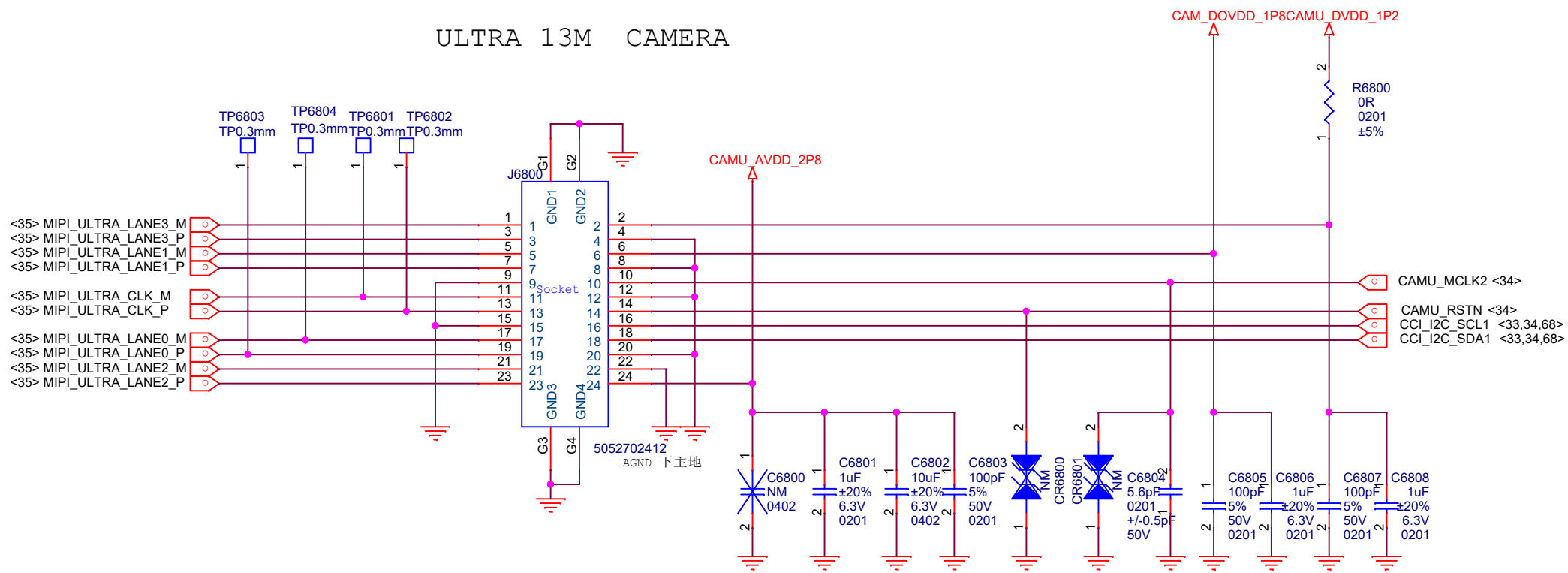
WIDE 64M

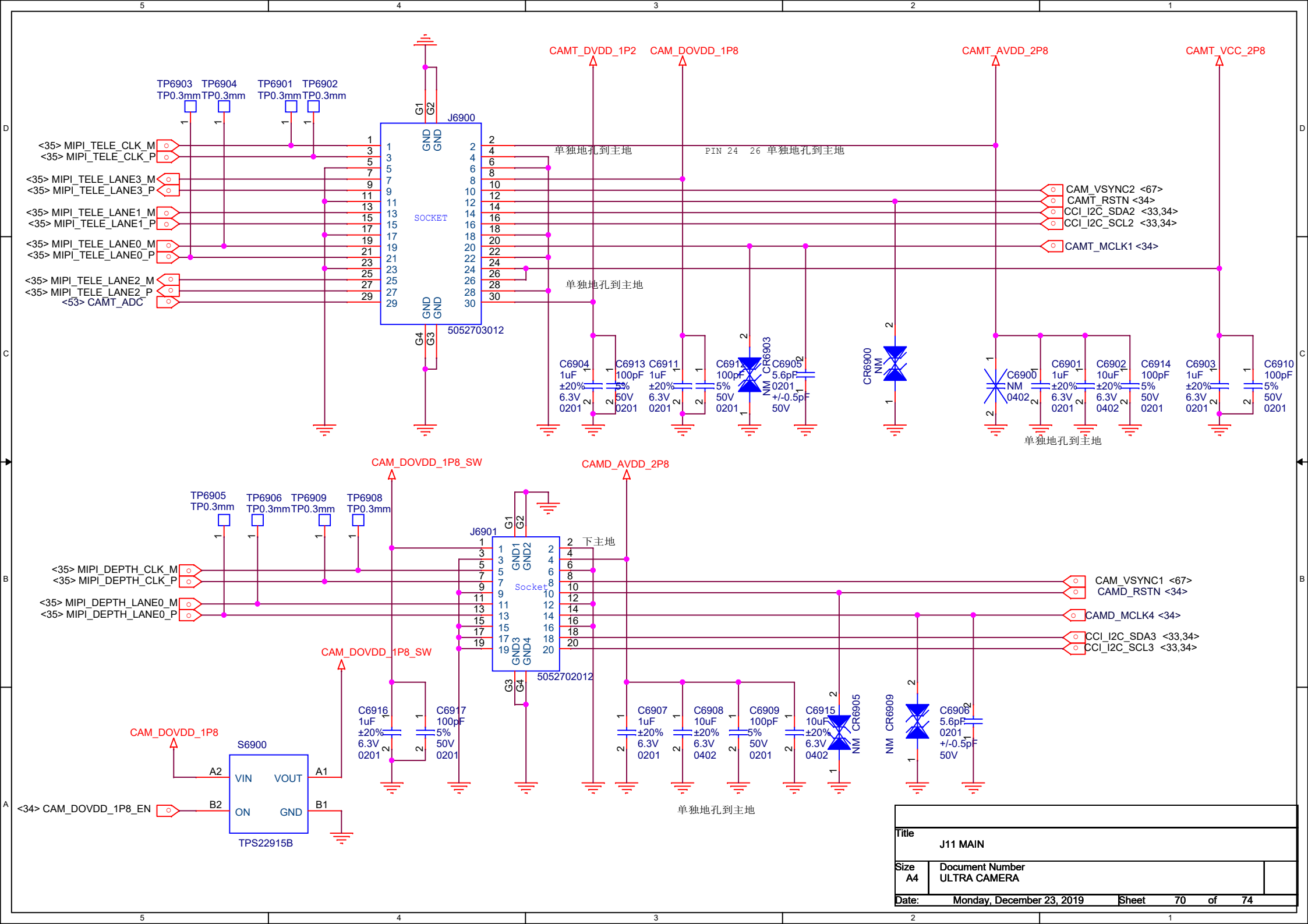


Title		
J11 MAIN		
Size	Document Number	
A4	WIDE&TELE CAMERA	
Date:	Monday, December 23, 2019	Sheet 67 74



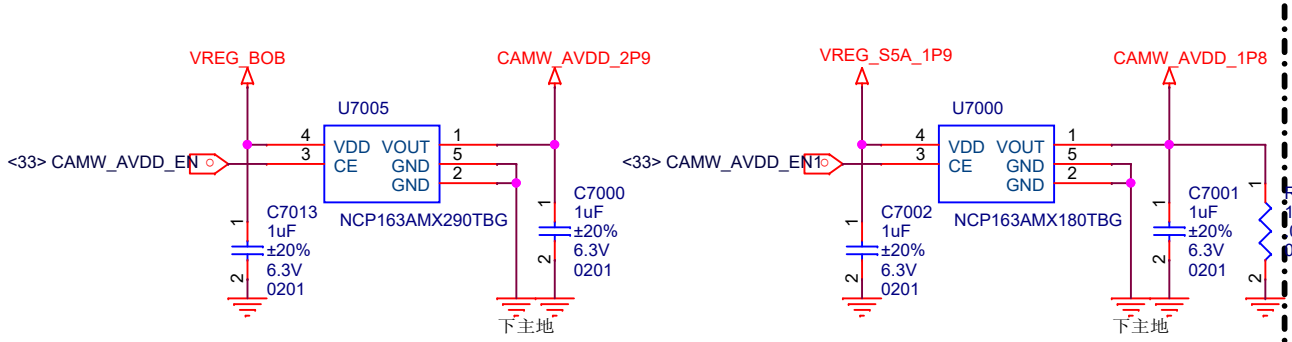
ULTRA 13M CAMERA



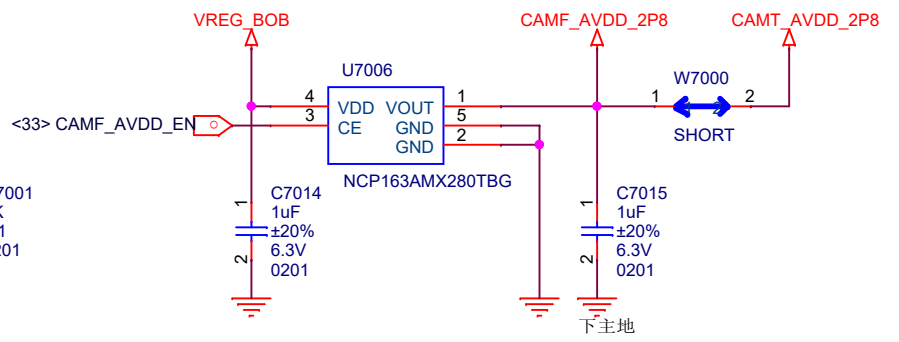


Title		
J11 MAIN		
Size	Document Number	
A4	ULTRA CAMERA	
Date:	Monday, December 23, 2019	Sheet 70 of 74

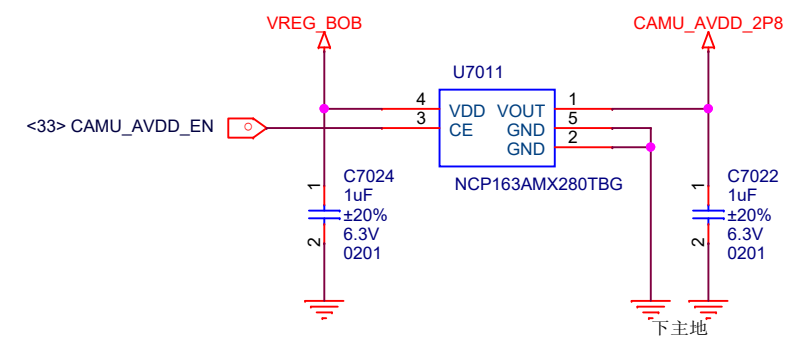
WIDE



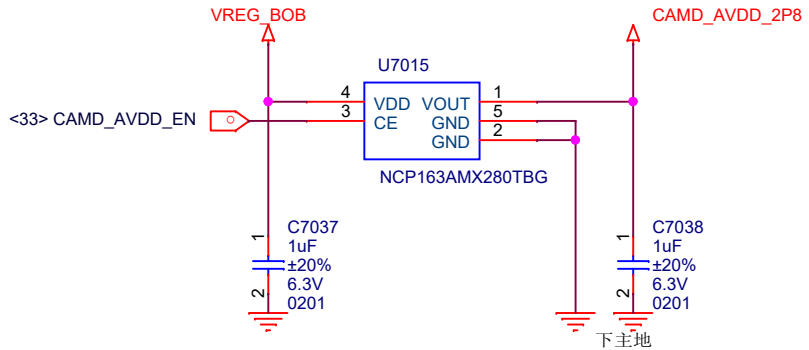
FRONT

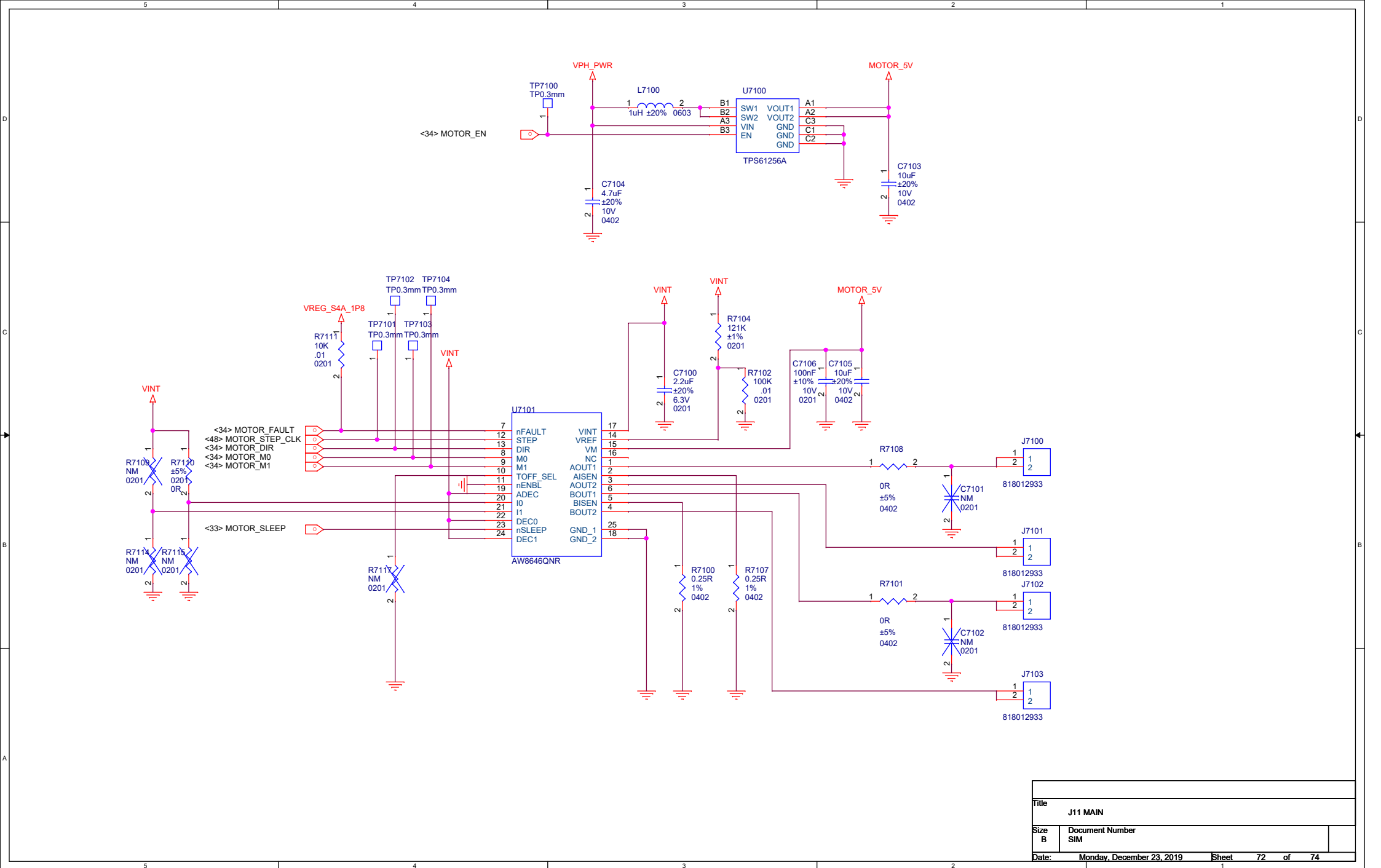


ULTRA

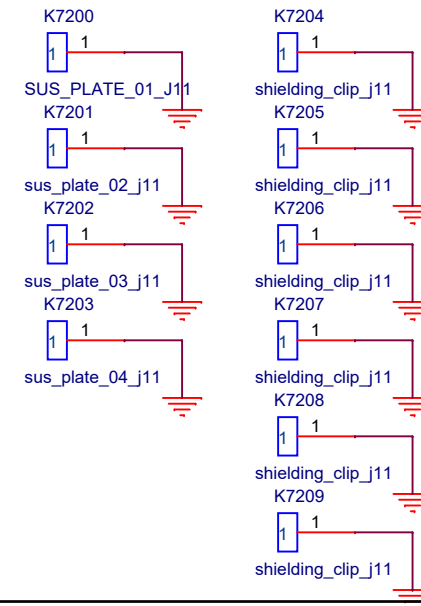
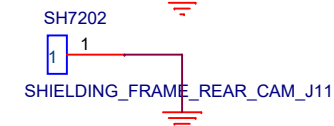
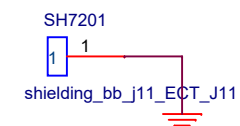
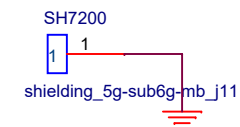
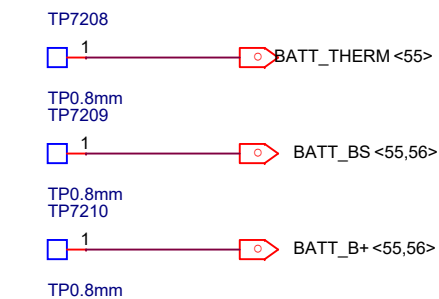
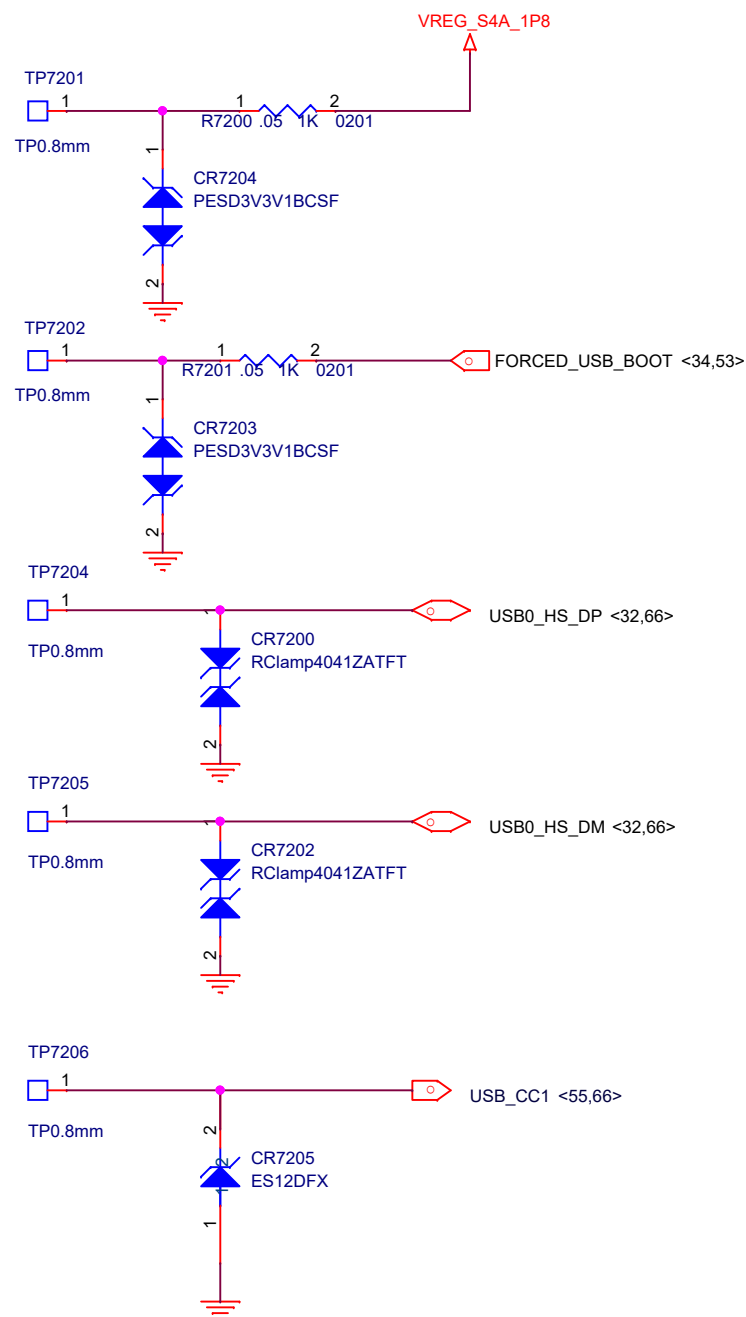
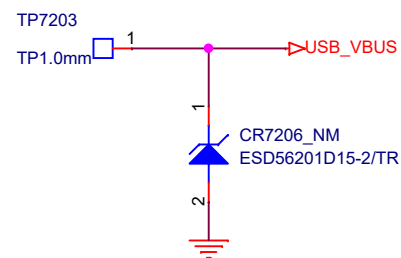
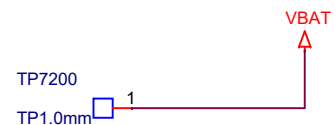


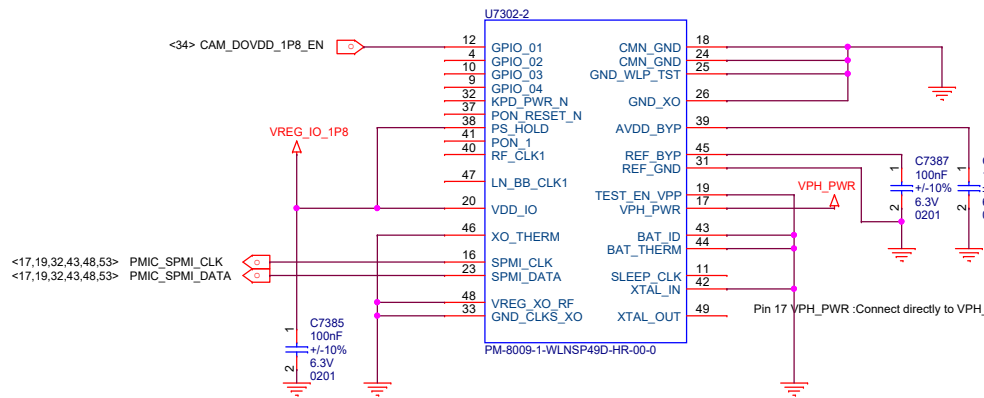
DEPTH





Title		
J11 MAIN		
Size	Document Number	
B	SIM	
Date:	Monday, December 23, 2019	Sheet 72 of 74





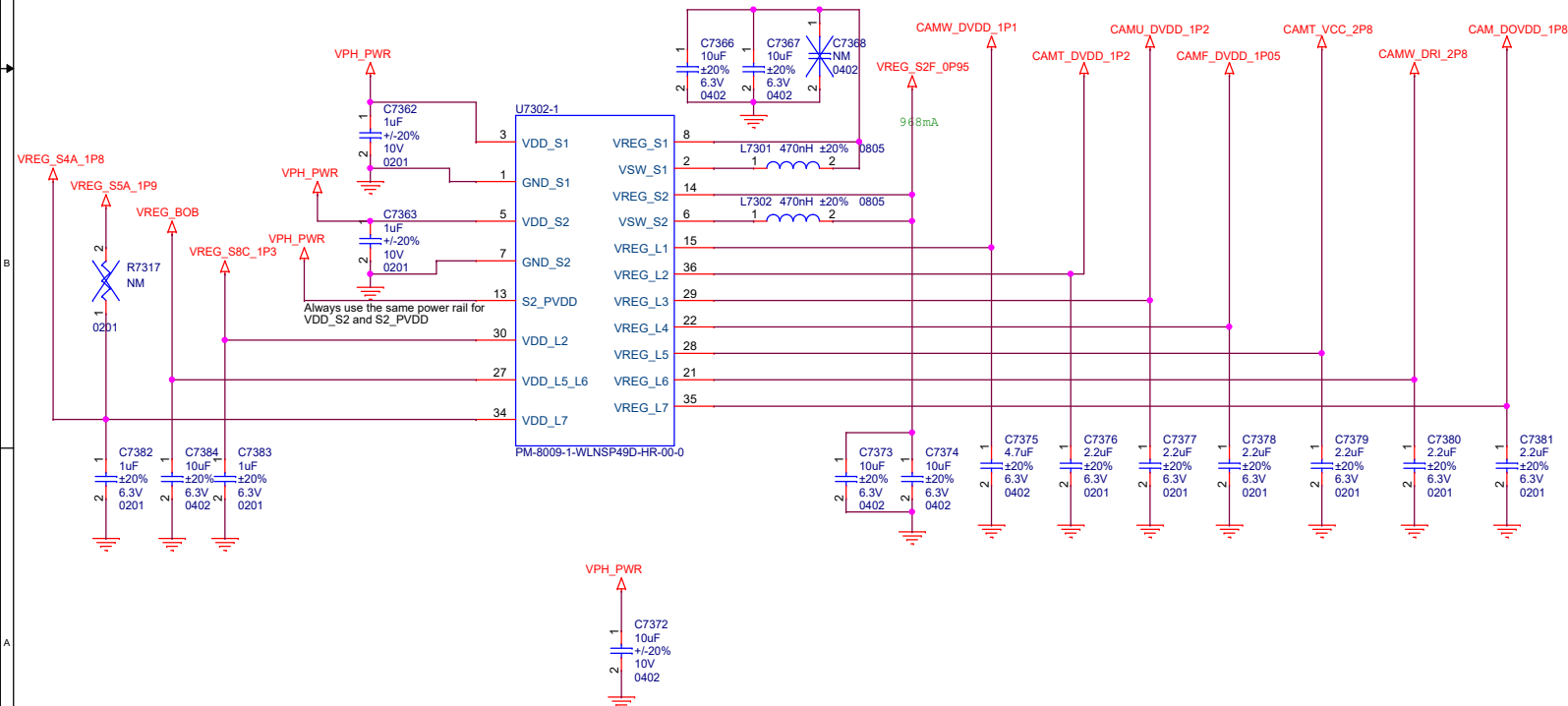
PM8009 AVDD_BYP should be routed away from noisy traces

Trace from pin to cap and ONE dedicated via from cap to main ground plane.
DCR < 200mOhm and ESL < 10nH
VPH Do NOT connect to any other ground.

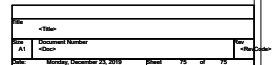
Dedicated trace from REF_BYP to cap.
Dedicated trace from REF_GND to cap.
Dedicated via to main ground plane right under PMIC pin (REF_GND) or as close to the pin as possible.
Do NOT add via at cap.

REF_BYP and REF_GND should be routed away from noisy traces

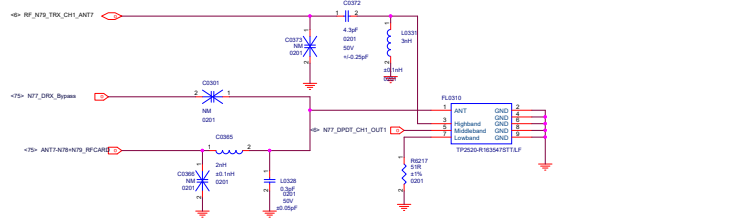
S1F will supply L1F, L4F, L3F INTERNALLY
Make sure VREG_S1 is routed with thick track as it carries VIN for L1F L3F L4F



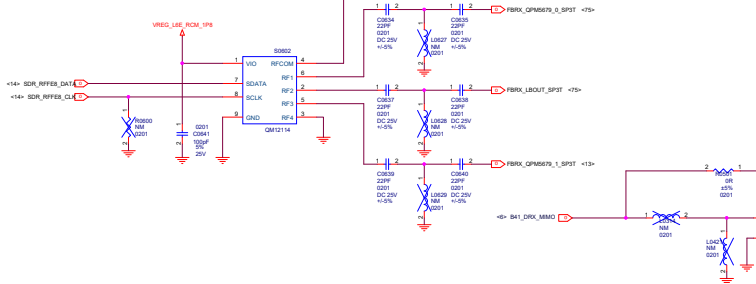
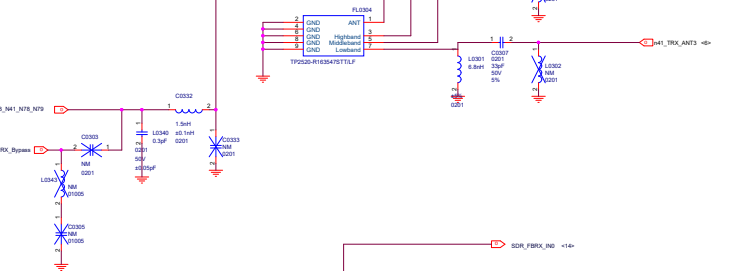
Title		F1 MAIN
Size	Document Number	CAMERA LDOs
Date:	Monday, December 23, 2019	Sheet 74



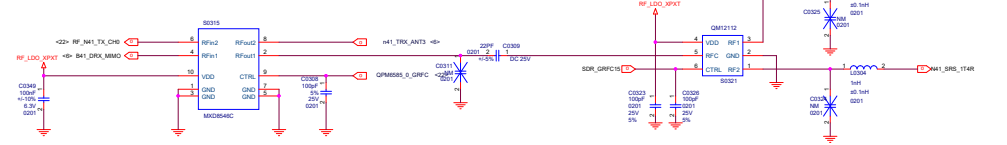
N78 N79 ANT7



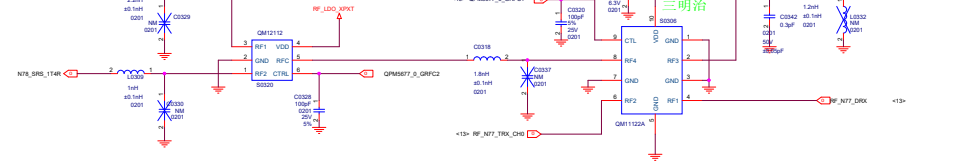
n41 TRX+n78 TRX+n79 TRX ANT3



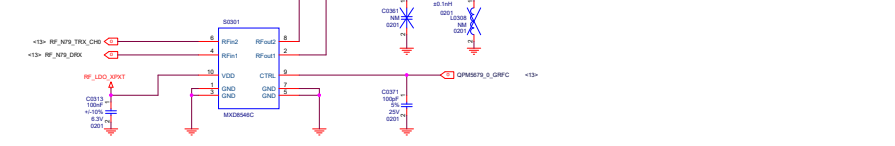
N41 DPDT1 三明治



n78 DPDT2 三明治

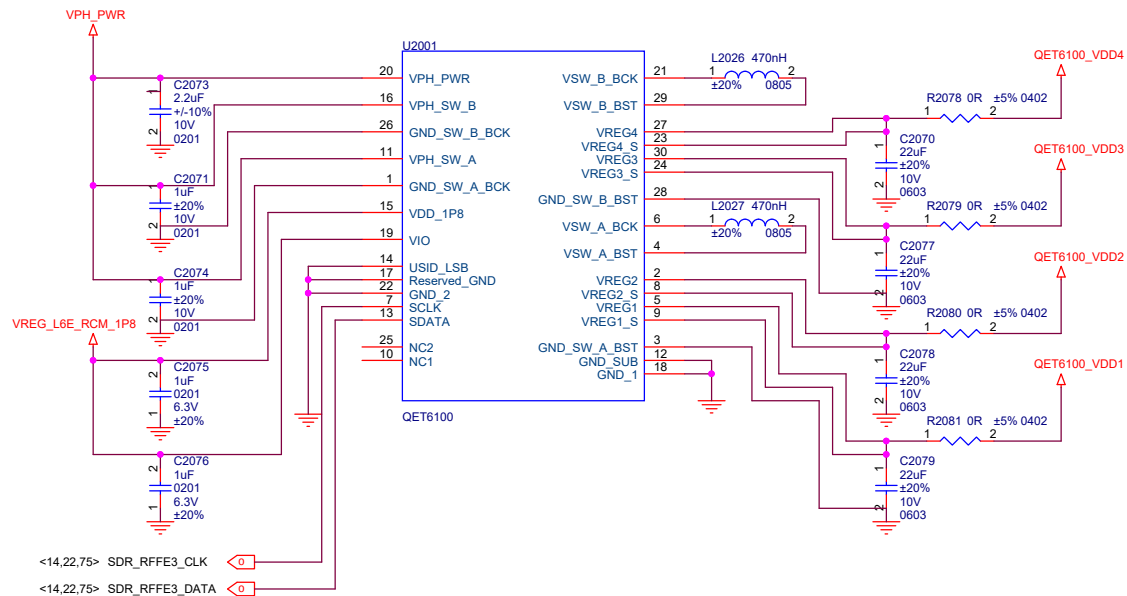
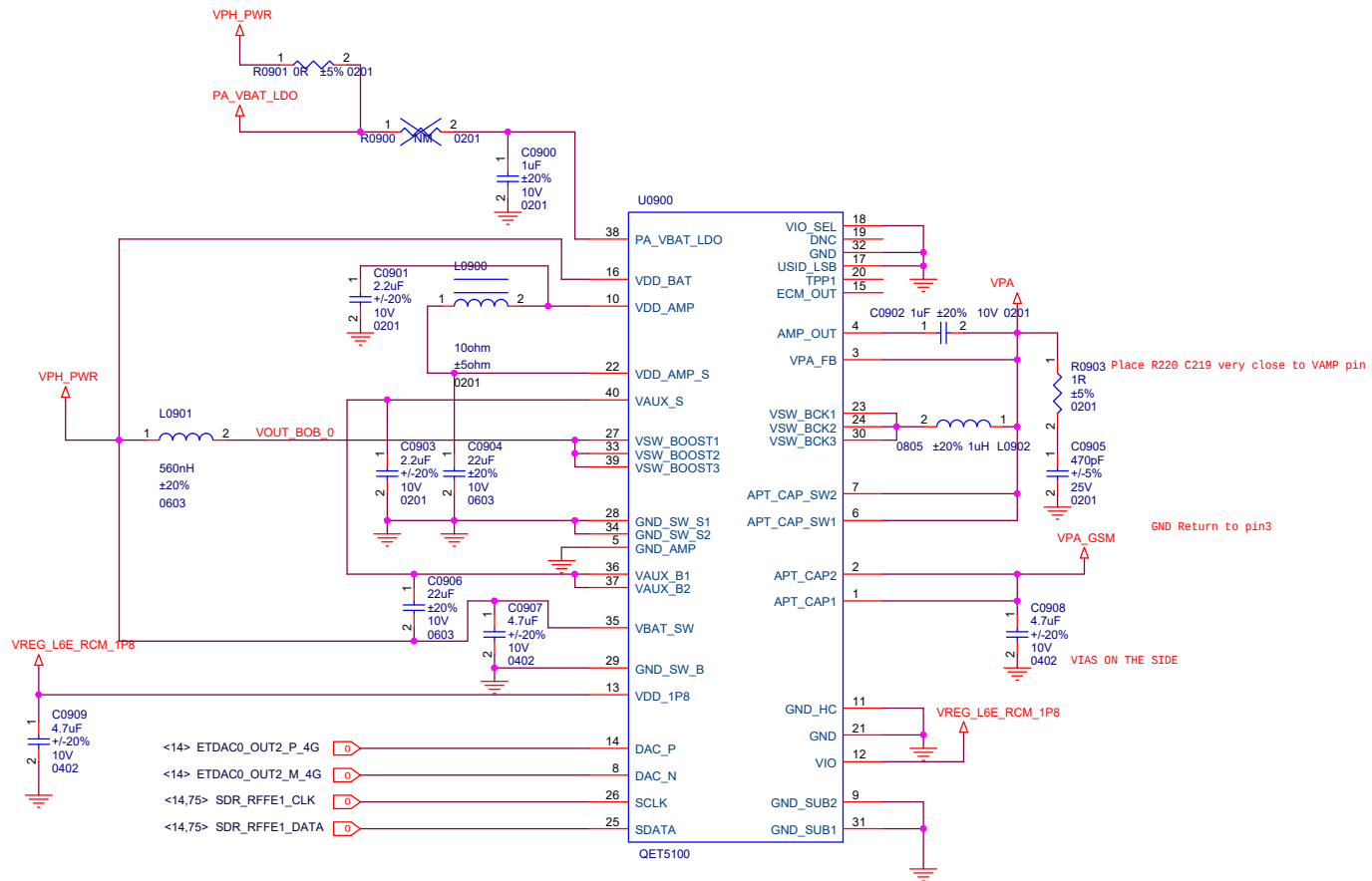


N79 DPDT1 三明治

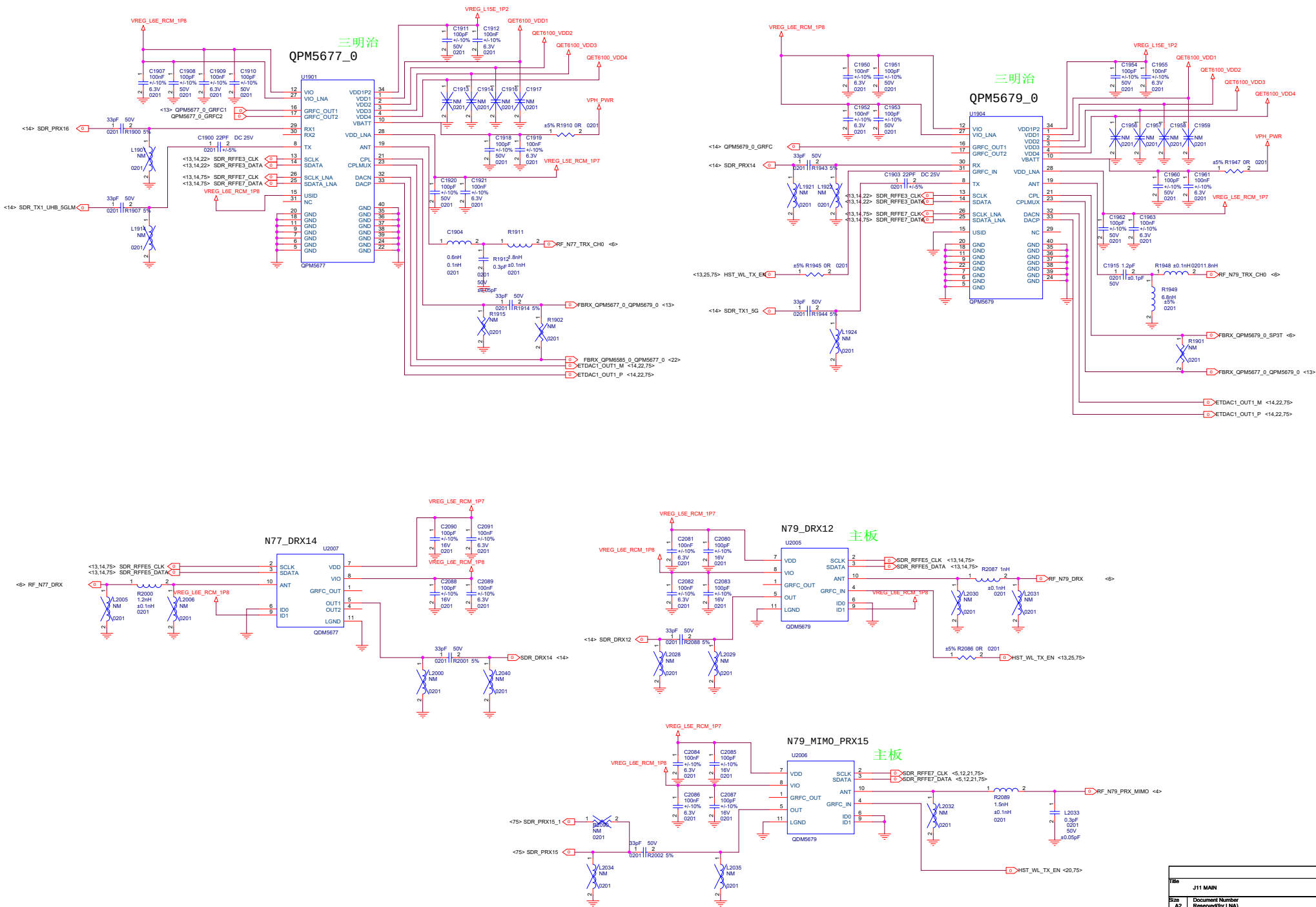


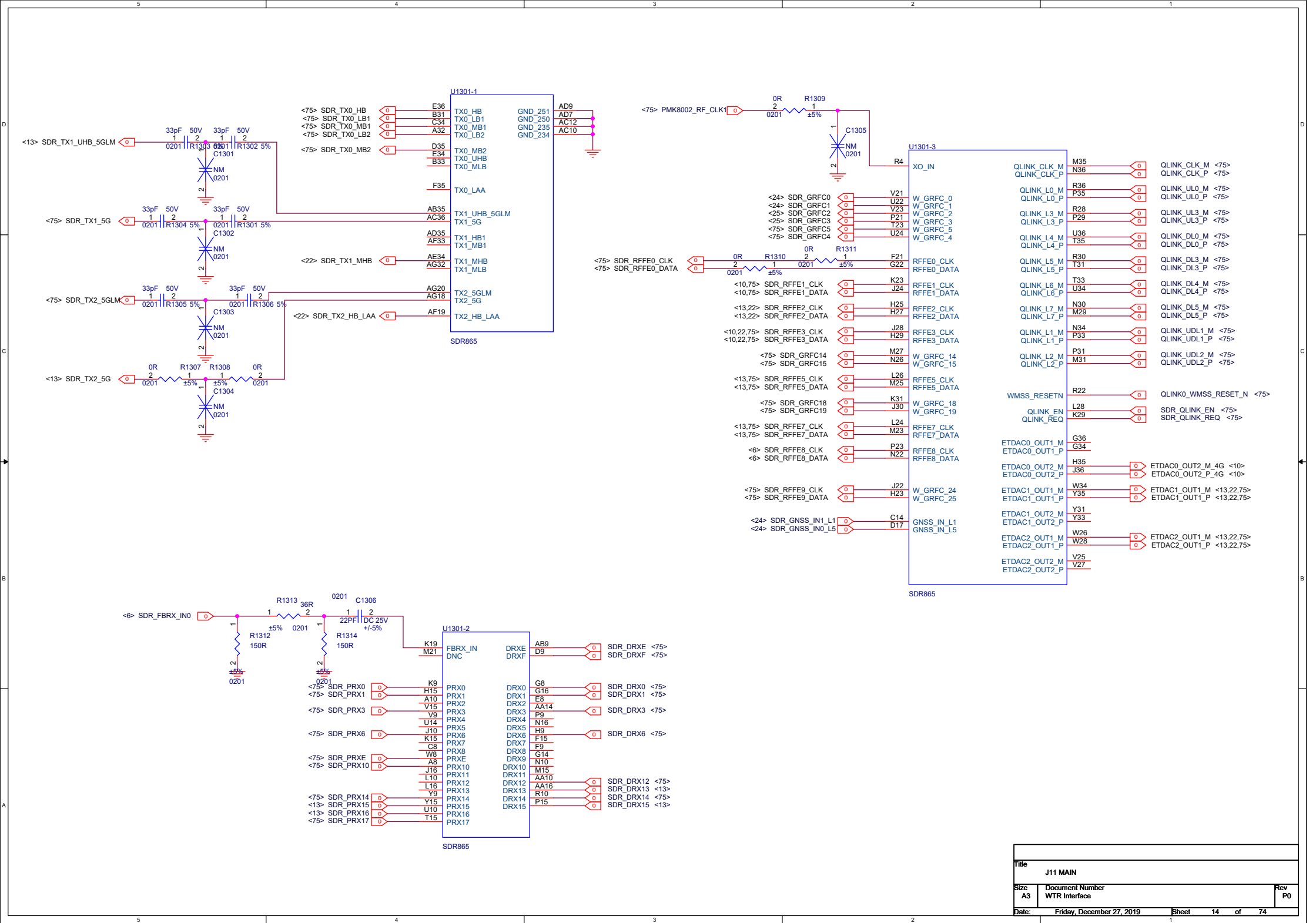
三明治

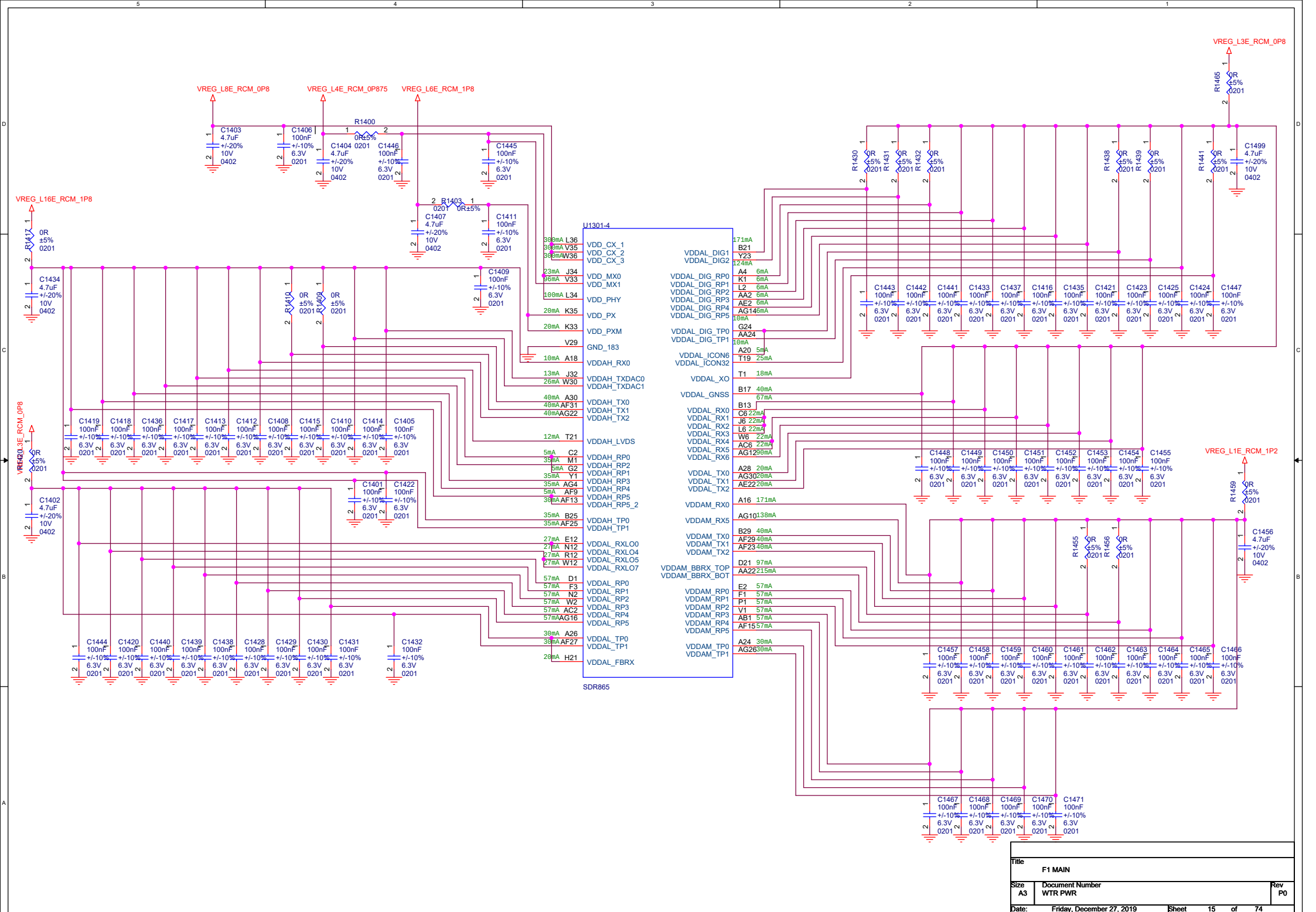




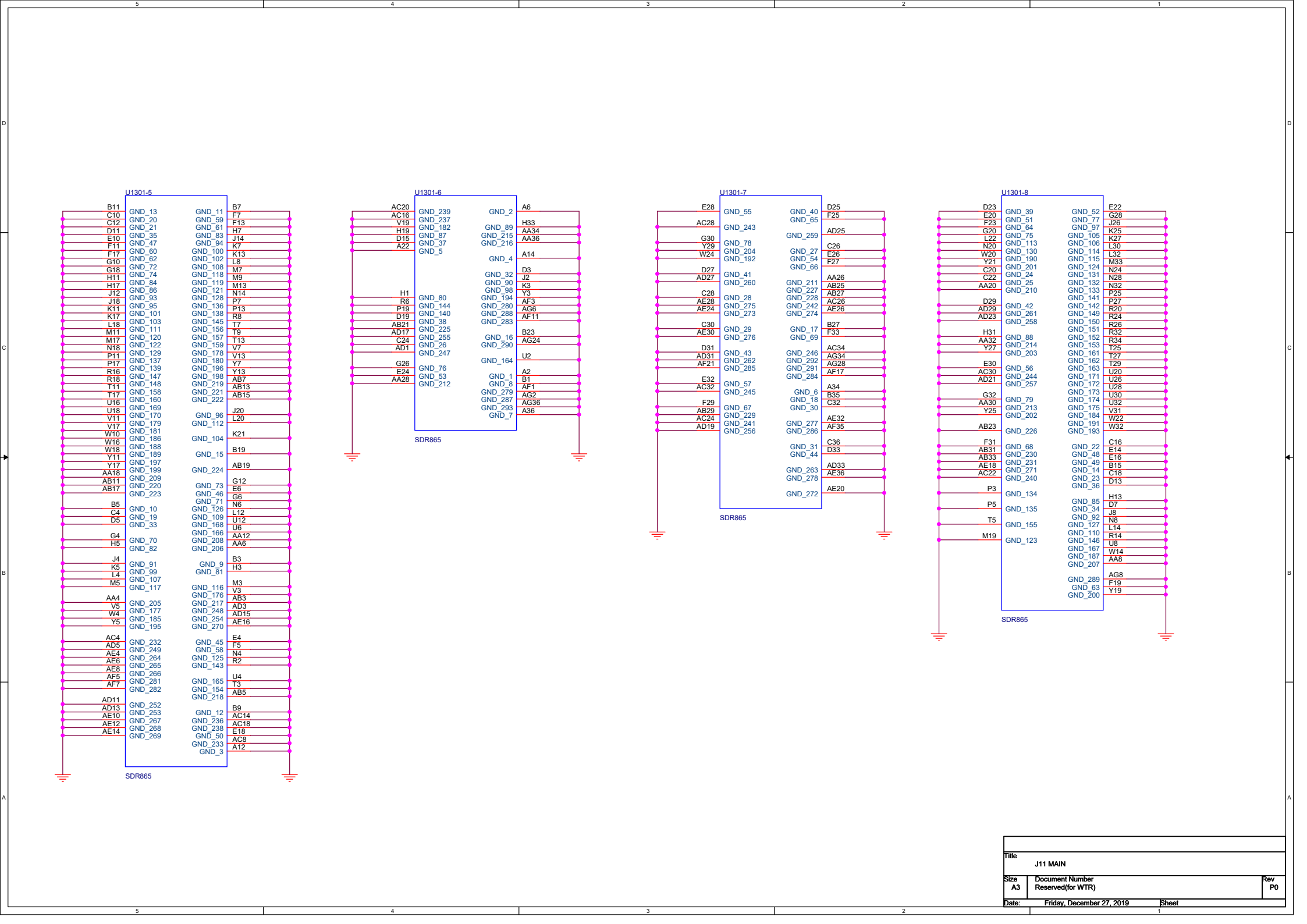
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J11 MAIN		
Size	Document Number	Rev
A3	RF DC-DC	P0
Date:	Friday, December 27, 2019	Sheet

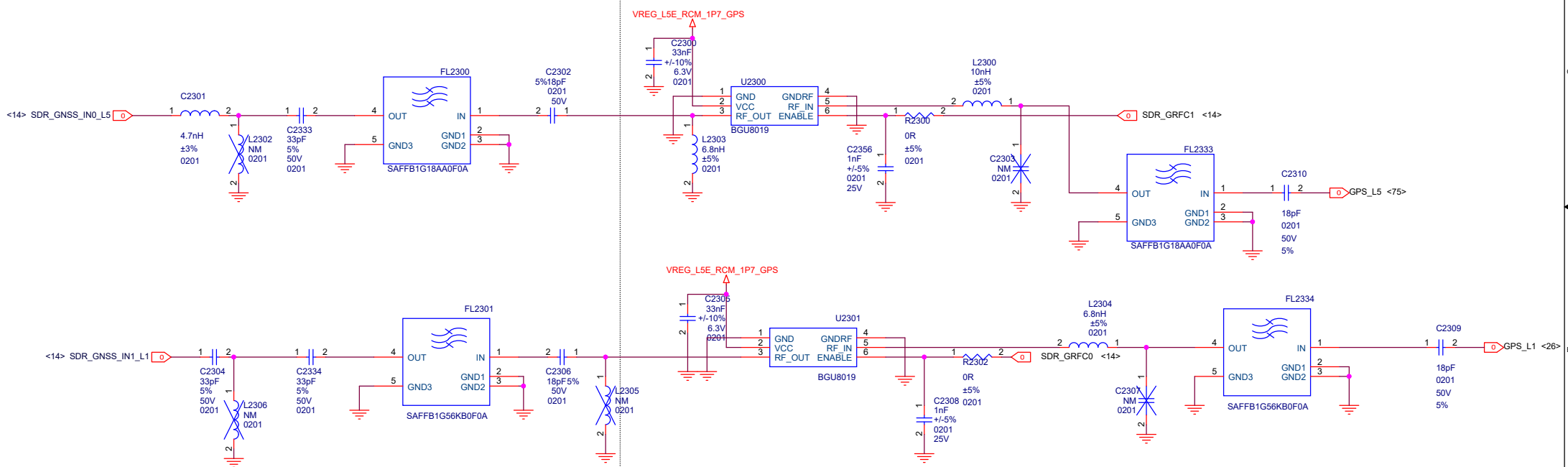






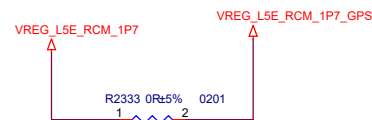
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Size			A3
Document Number			WTR PWR
Rev			P0
Date:			Friday, December 27, 2019
Sheet			15 of 74





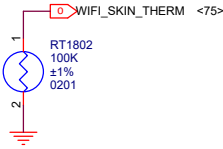
Close2WTR

Close2WCN

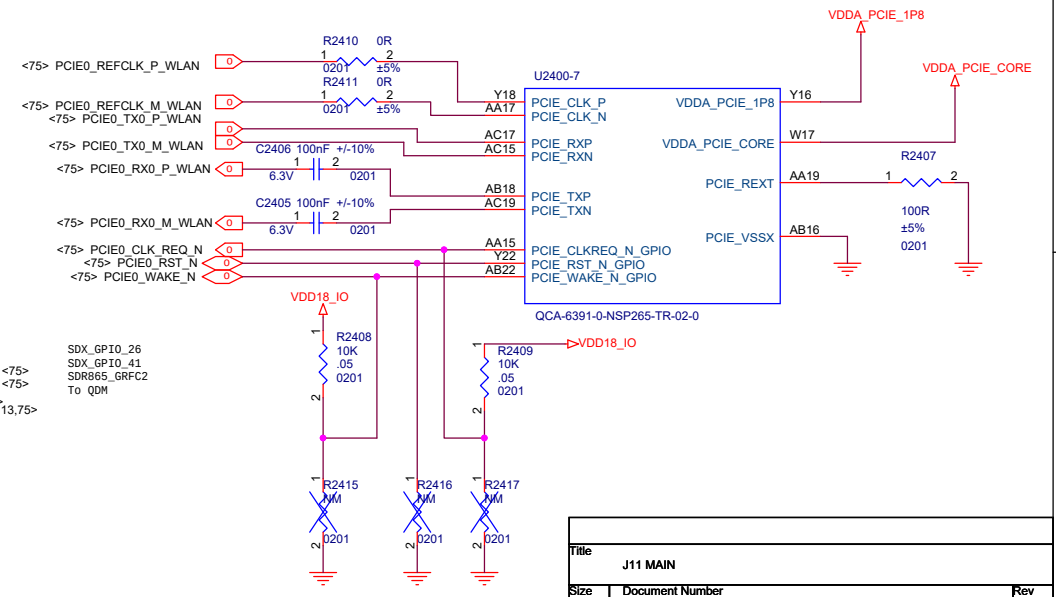
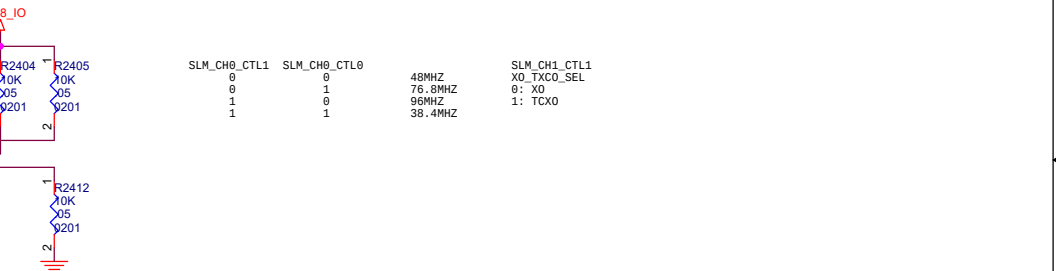
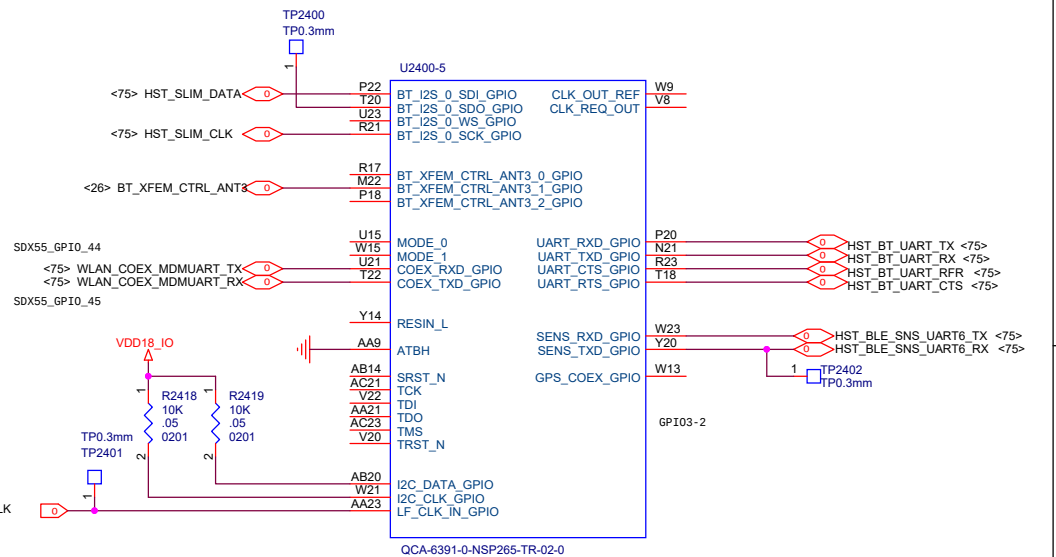
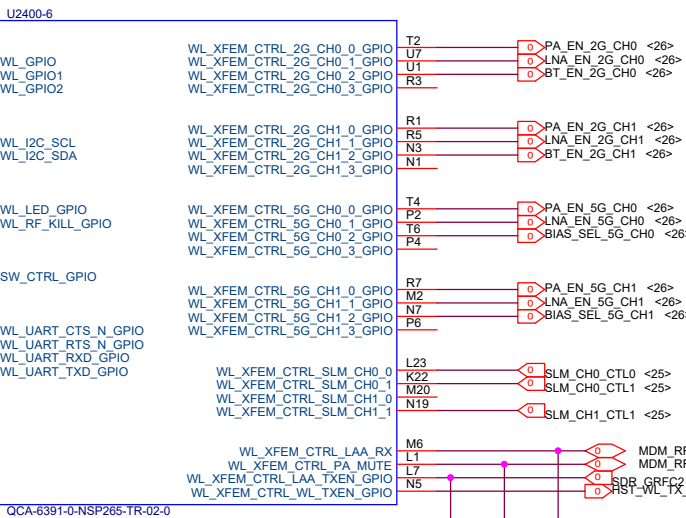
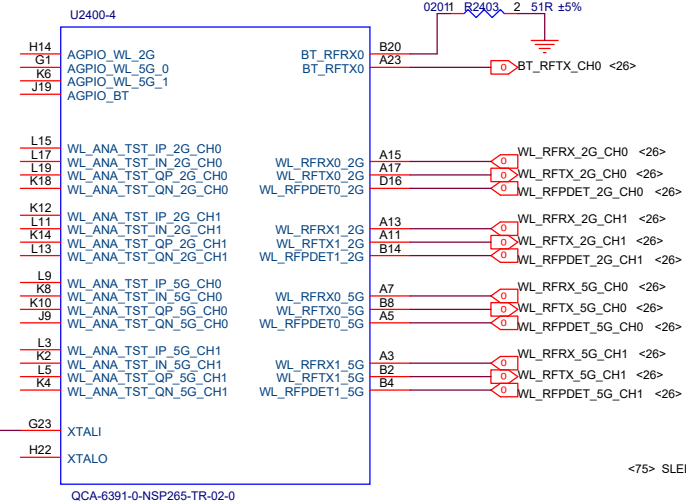
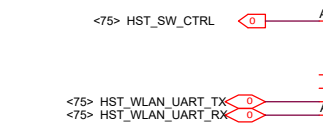
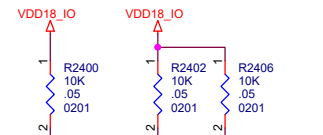
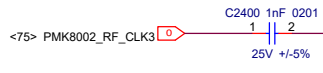


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J11 MAIN		
Size	Document Number	Rev
A3	GPS	P0
Date:	Friday, December 27, 2019	Sheet

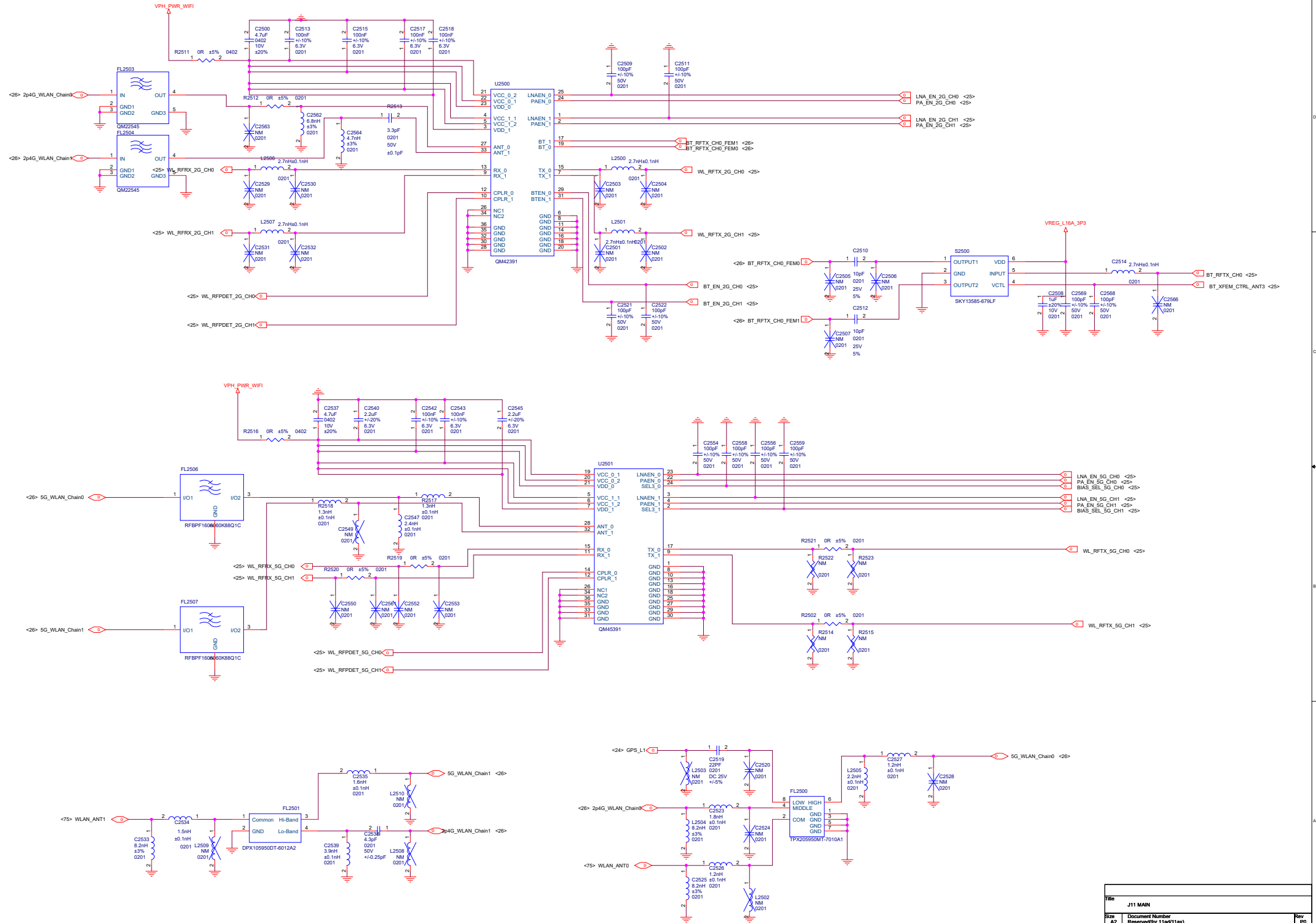
放U2400旁边



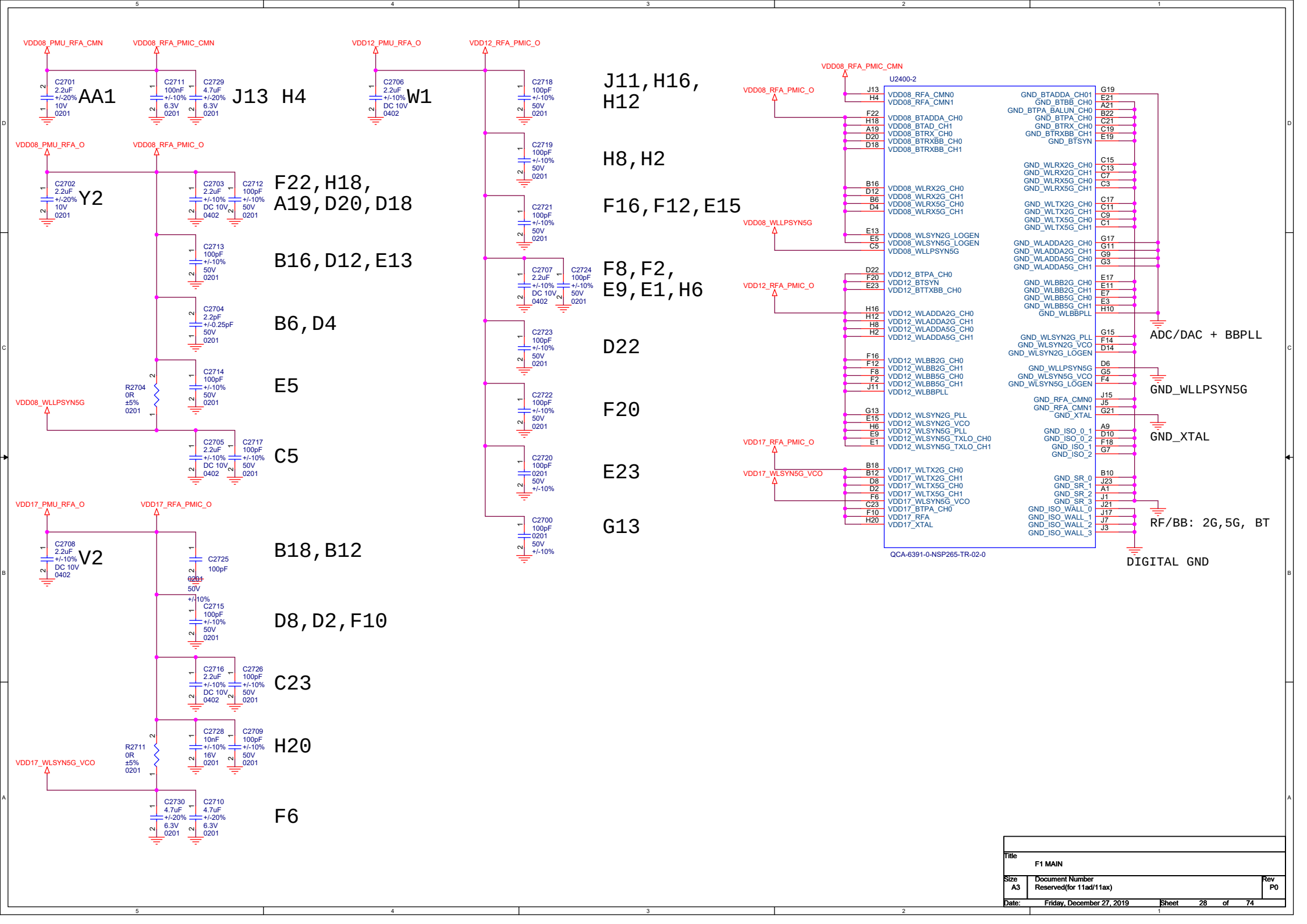
Overlap one pad of R2401 and C2403
Overlap one pad of R2409 and C2404
Overlap one pad of C2403 and C2404



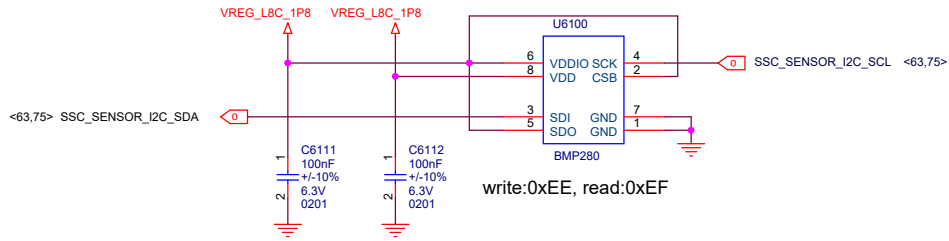
Title			J11 MAIN
Size	A3	Document Number	WIFI Interface
Date:	Friday, December 27, 2019	Sheet	25 of 74



File			J11 MAIN
Size	A2	Document Number	Reserved (for 11a/11ac)
Date	Friday, December 27, 2019	Sheet	8

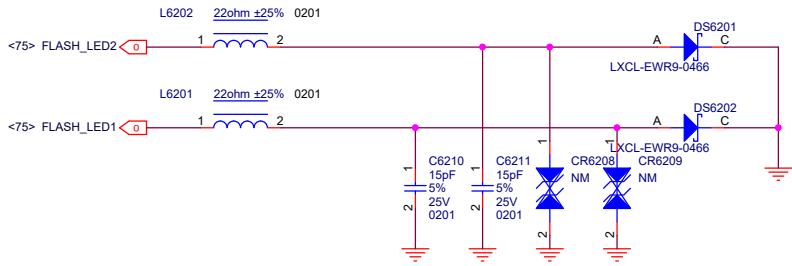


PRESSURE

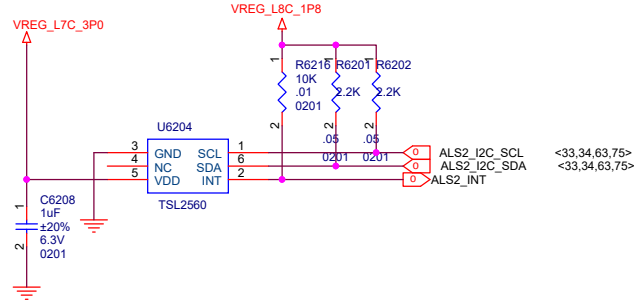


FLICKER

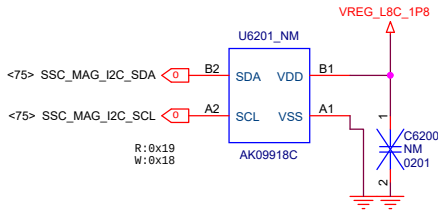
FLASH



ALS



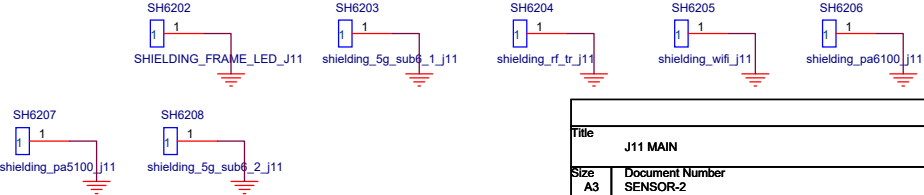
COMPASS



GND Testpoint



Shielding



Title		
J11 MAIN		
Size	Document Number	Rev
A3	SENSOR-2	P0
Date:	Friday, December 27, 2019	Sheet 63 of 74

